PipeCheck: Specifying and Verifying Microarchitectural Enforcement of Memory Consistency Models

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and Margaret Martonosi[+]

[+]Princeton University [*]Intel VSSAD

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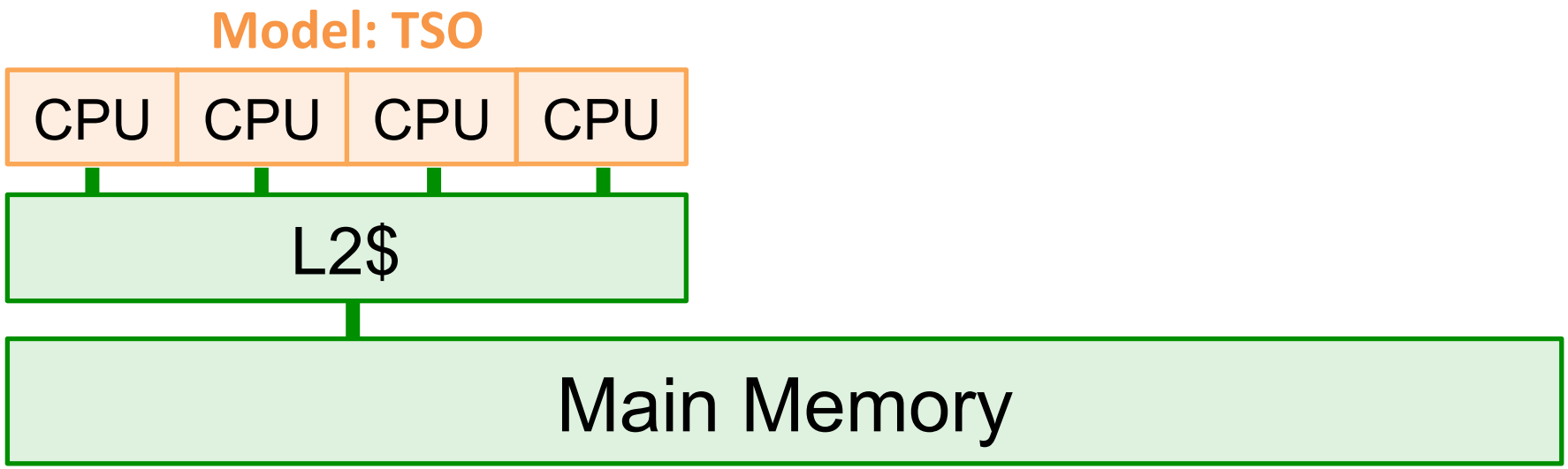
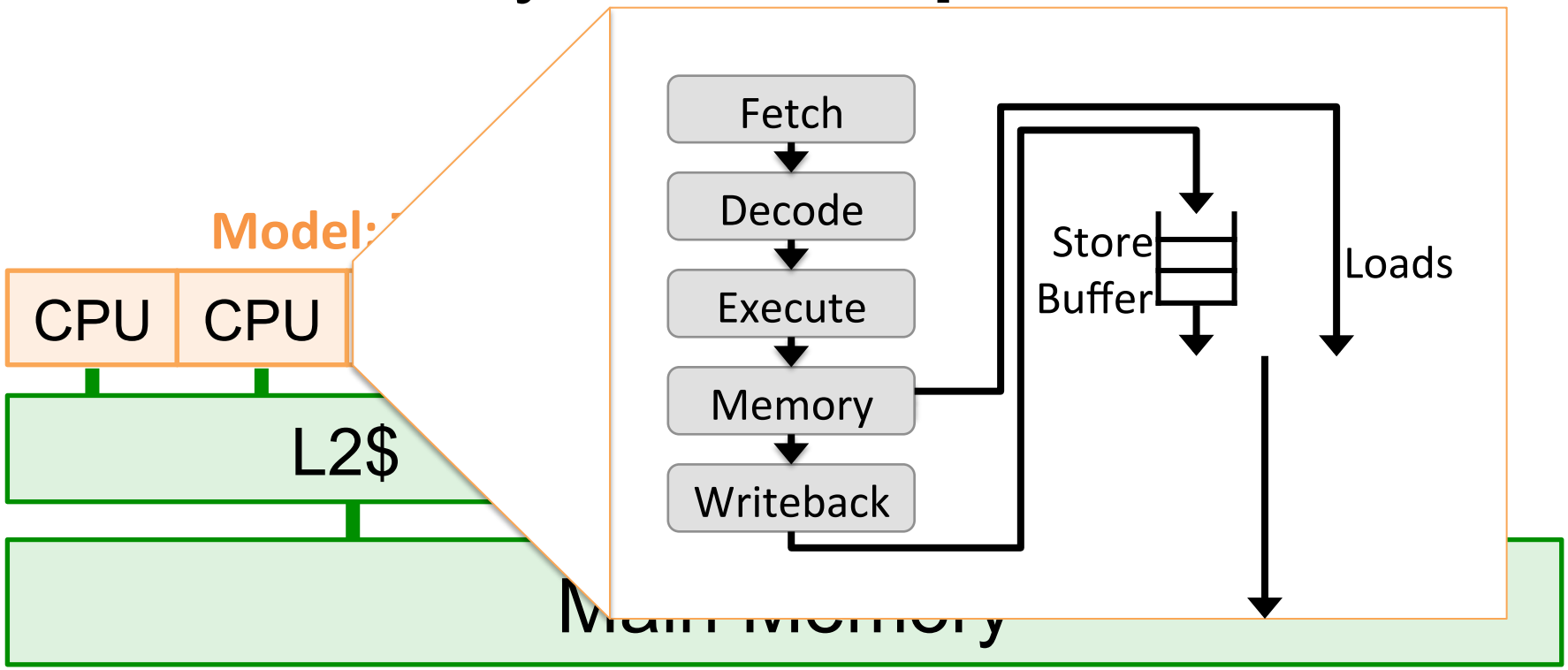
PRINCETON

Multicore Systems Need Memory Consistency Models



Motivation: Verify correctness of memory consistency model **implementation**



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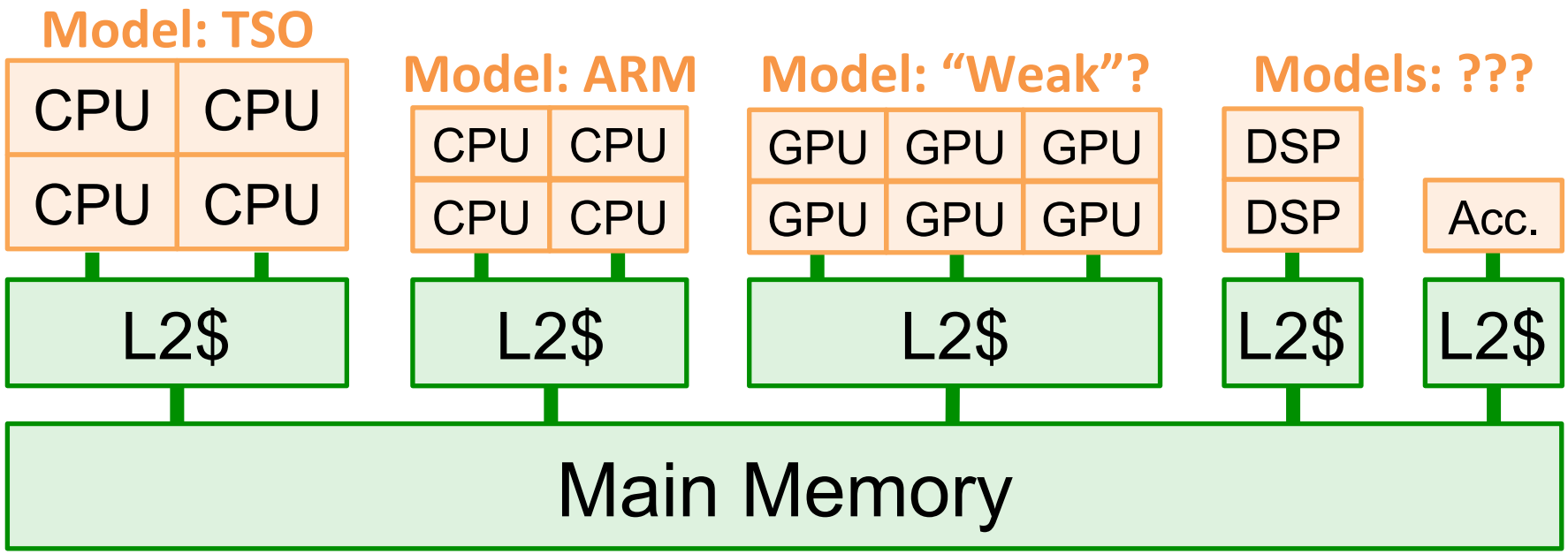

- Systems are getting more complicated!

Talk Outline

- Architectural Models: Defining Correctness
- PipeCheck µarch-Level Analysis
  - Basic "µarch-happens-before" Graphs
  - Advanced µarch Optimizations
  - Comparing arch and µarch
- Automated Tool and Performance Results
- Conclusions

Memory Models Defined by ISA

8.2.2 Memory Ordering in P6 and More Recent Processor Families

The Intel Core 2 Duo, Intel Atom, Intel Core Duo, Pentium 4, and P6 family processors also use a processor-ordered memory-ordering model that can be further defined as "write ordered with store-buffer forwarding." This model can be characterized as follows.

In a single-processor system for memory regions defined as write-back cacheable, the memory-ordering model respects the following principles (**Note** the memory-ordering principles for single-processor and multiple-processor systems are written from the perspective of software executing on the processor, where the term "processor" refers to a logical processor. For example, a physical processor supporting multiple cores and/or HyperThreading Technology is treated as a multi-processor systems.):

- Reads are not reordered with other reads.
- Writes are not reordered with older reads.
- Writes to memory are not reordered with other writes, with the following exceptions:
  - writes executed with the CLFLUSH instruction;
  - streaming stores (writes) executed with the non-temporal move instructions (MOVNTI, MOVNTQ, MOVNTDQ, MOVNTPS, and MOVNTPD); and
  - string operations (see Section 8.2.4.1).
- Reads may be reordered with older writes to different locations but not with older writes to the same location.

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Type A (first)

Type B (second)

	Load	Store
Load	No	No
Store	Yes	No

...

Lustig, Pellauer, Martonosi, "PipeCheck"

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How to automate verification of microarchitectural implementations?

Type A (first)

Type B (second)

	Load	Store
Load	No	No
Store	Yes	No

What is the Definition of "Correct"?

Preserved Program Order (PPO) for x86-TSO: Reorderings Permitted?

Type A (first) ↑ Type B (second)

	Load	Store
Load	No	No
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[Intel, AMD, Alglave, FMSD '12, Owens et al., TPHOLs '09]

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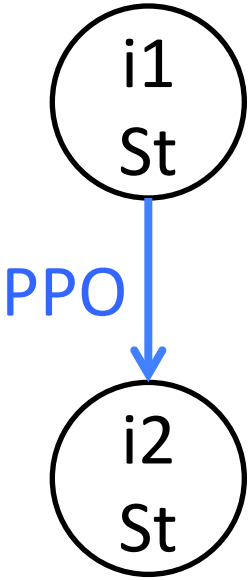

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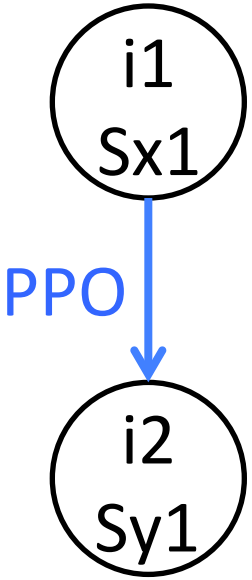
Initially: [x]=[y]=0

Core 0	Core 1
(i1) st [x], 1 (i2) st [y], 1	(i3) ld [y] → r1 (i4) ld [x] → r2

Can core 1 observe (i2) before it observes (i1)?



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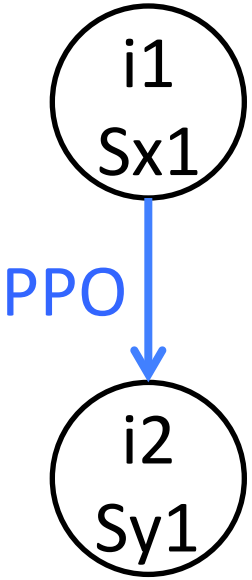

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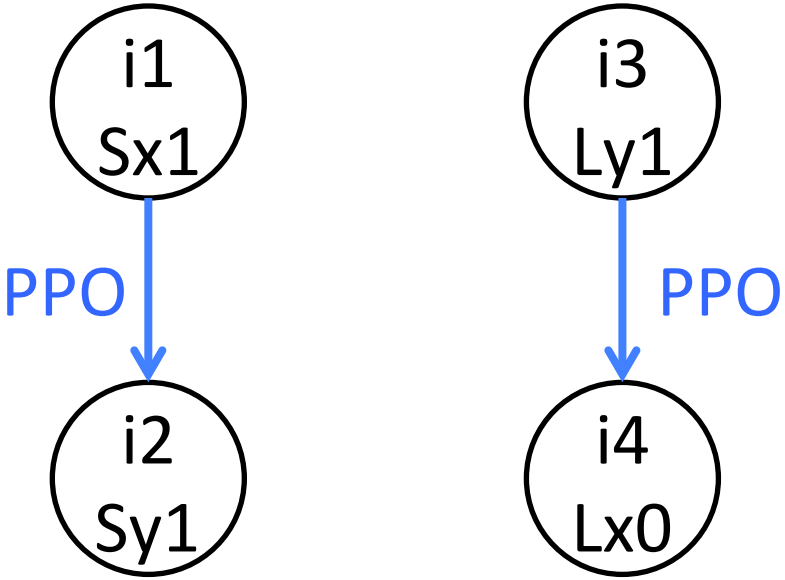

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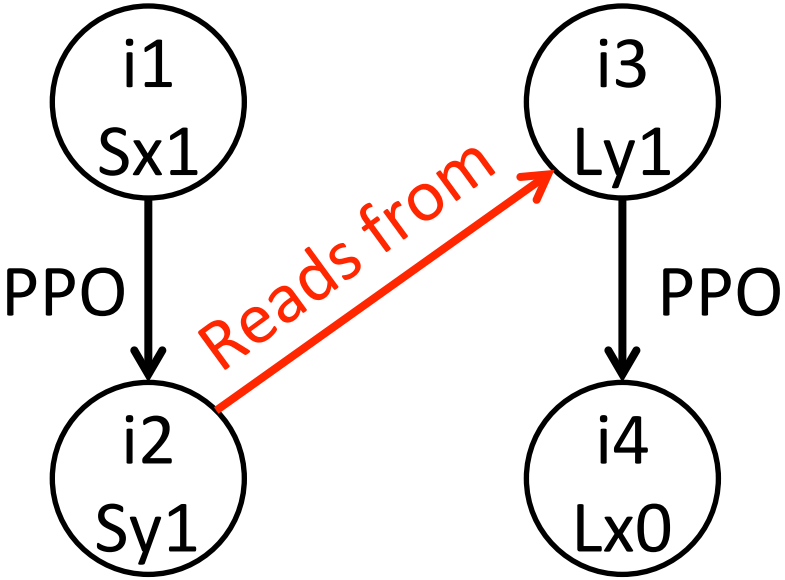
r1=1



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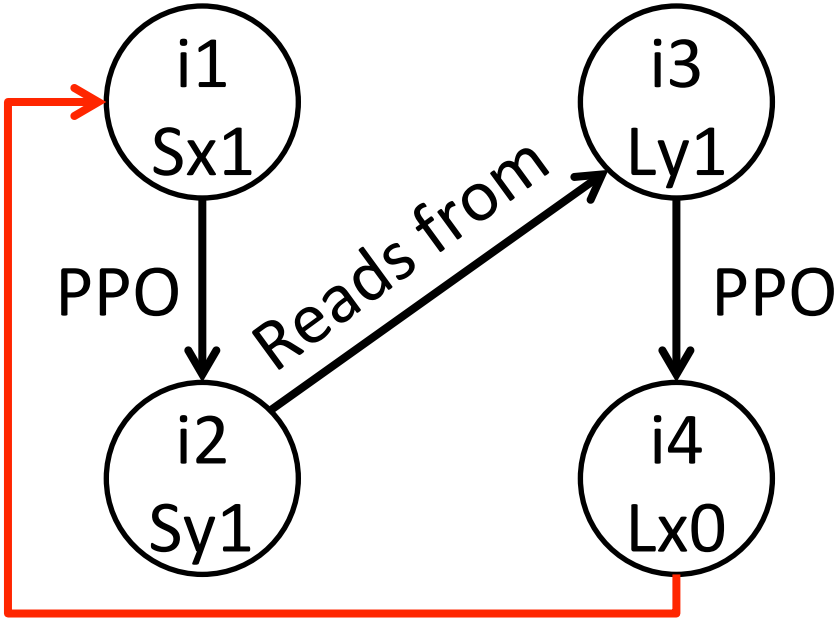
r1=1, **r2=0**



Reads from earlier store ("from-reads")

[Intel, AMD, Alglave, FMSD '12, Owens et al., TPHOLs '09]

What is the Definition of "Correct"?

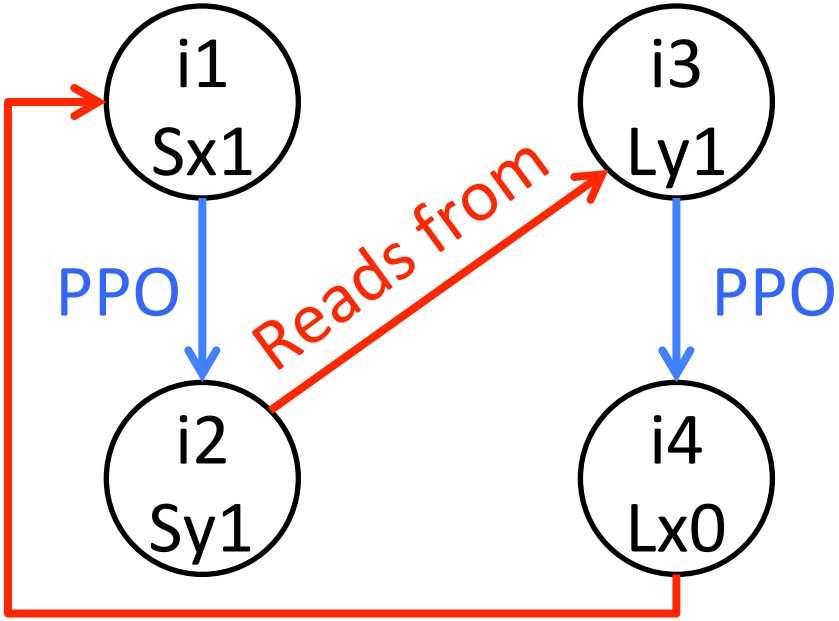
Litmus test mp:

Initially: [x]=[y]=0

Core 0	Core 1
(i1) st [x], 1 (i2) st [y], 1	(i3) ld [y] → r1 (i4) ld [x] → r2

TSO: **Forbid**: r1=1, r2=0

Cycle implies forbidden execution: instruction can't happen before itself



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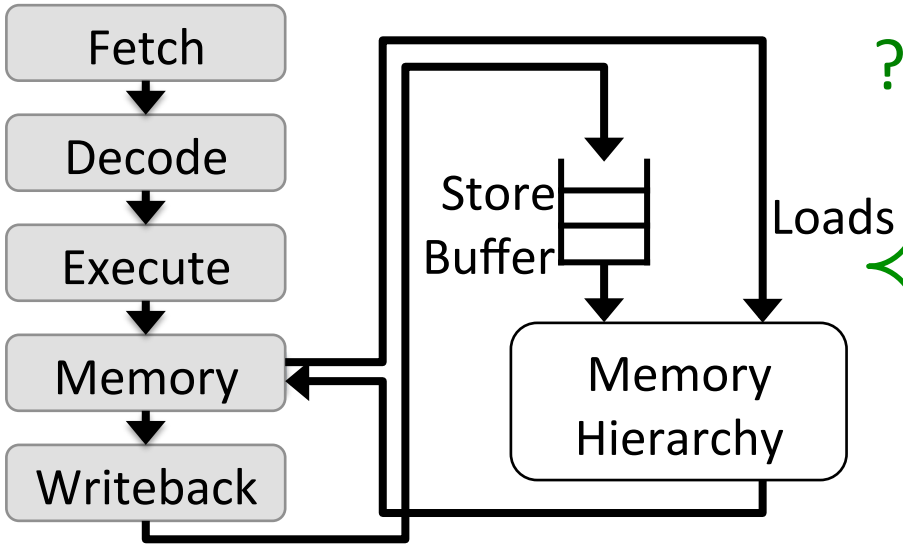
How is the consistency model **enforced at the microarchitecture level**?



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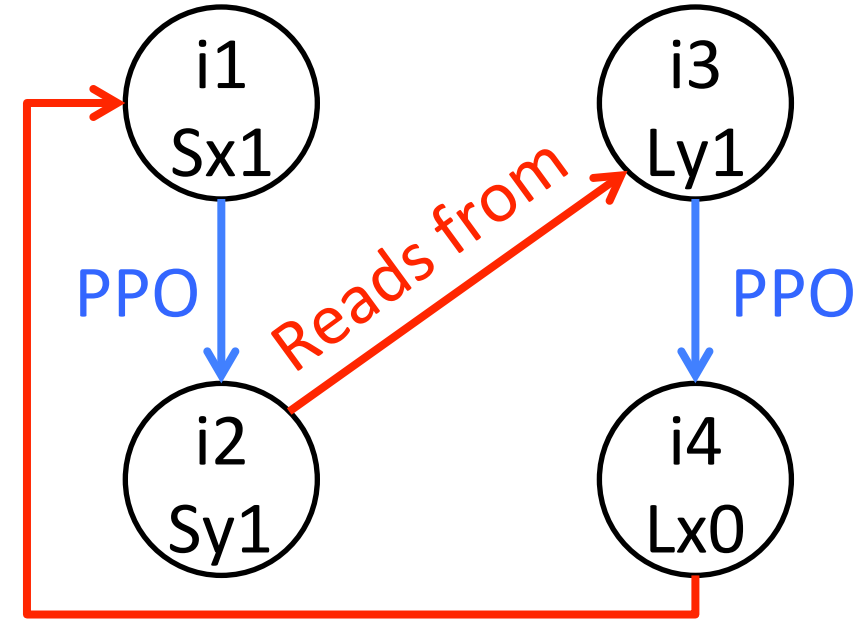

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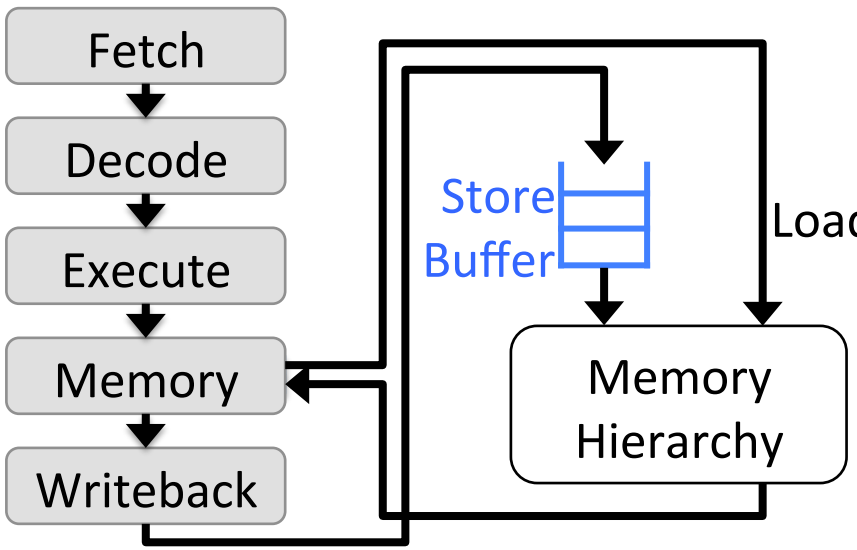

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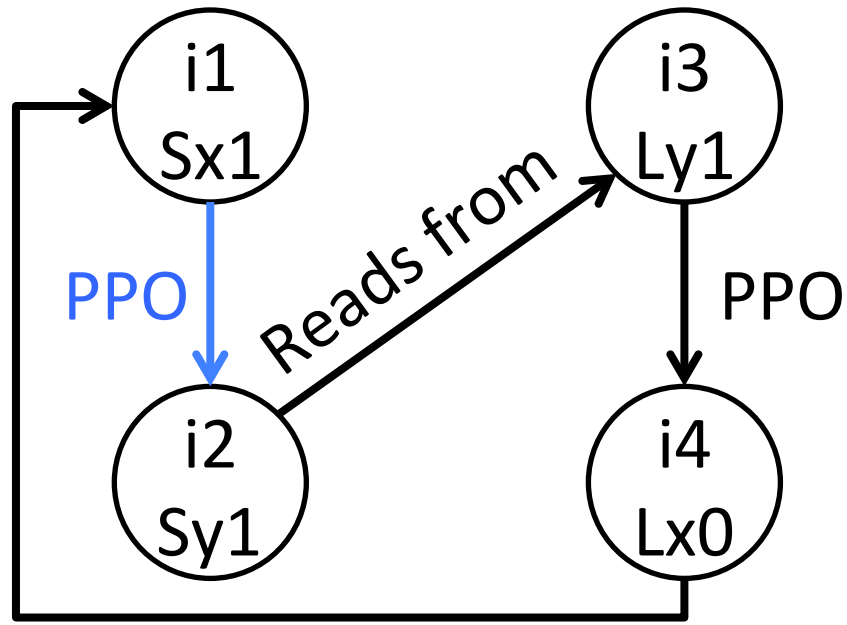

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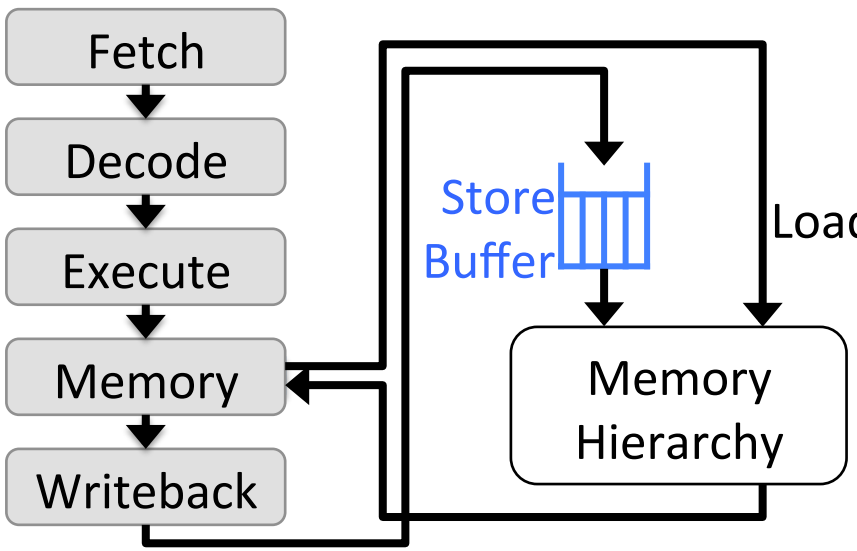

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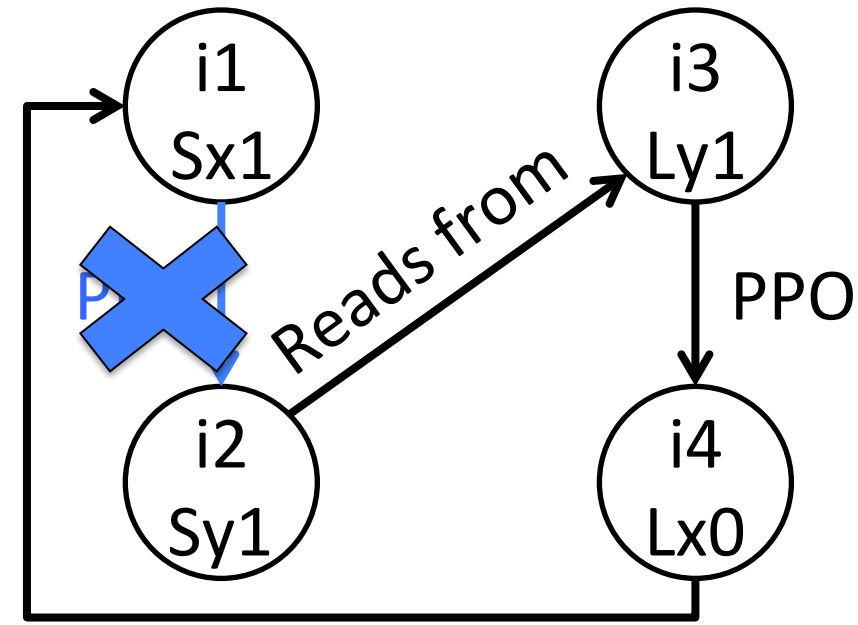

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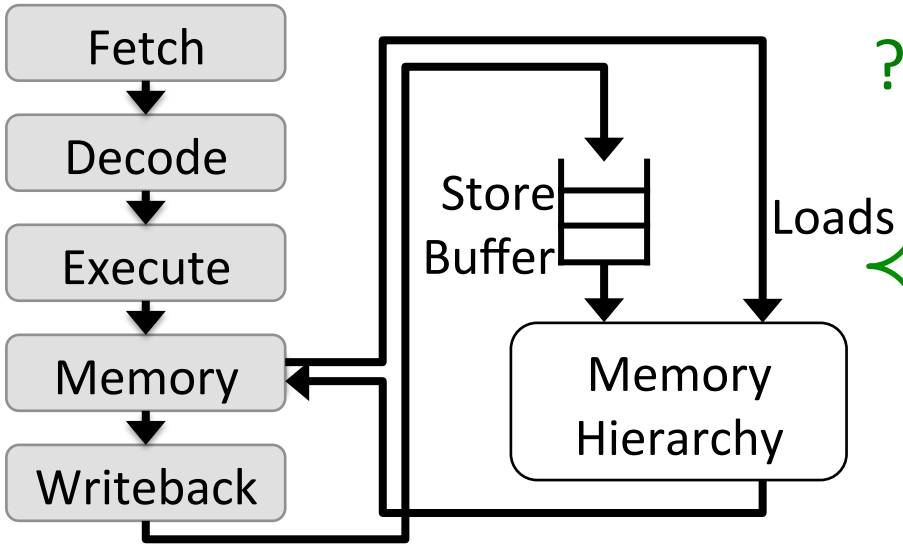

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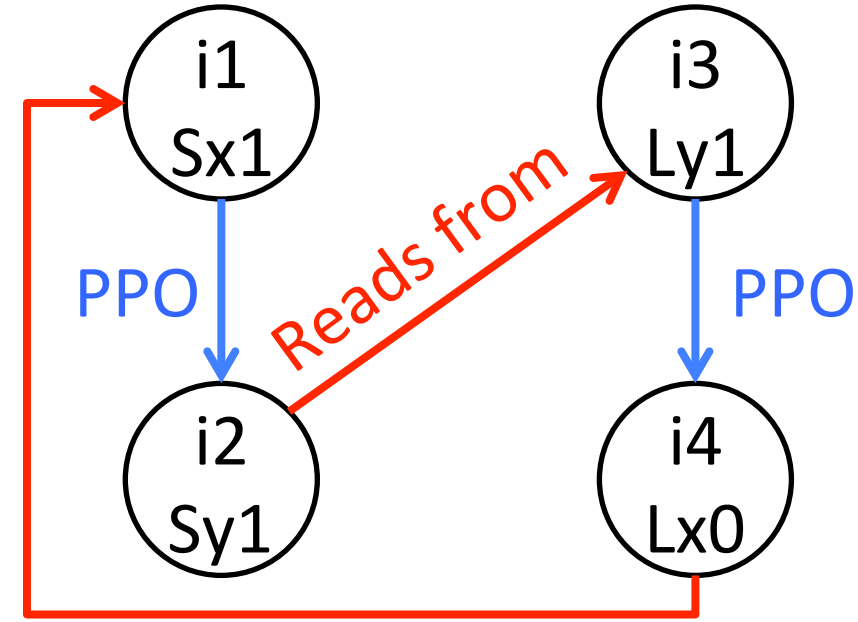

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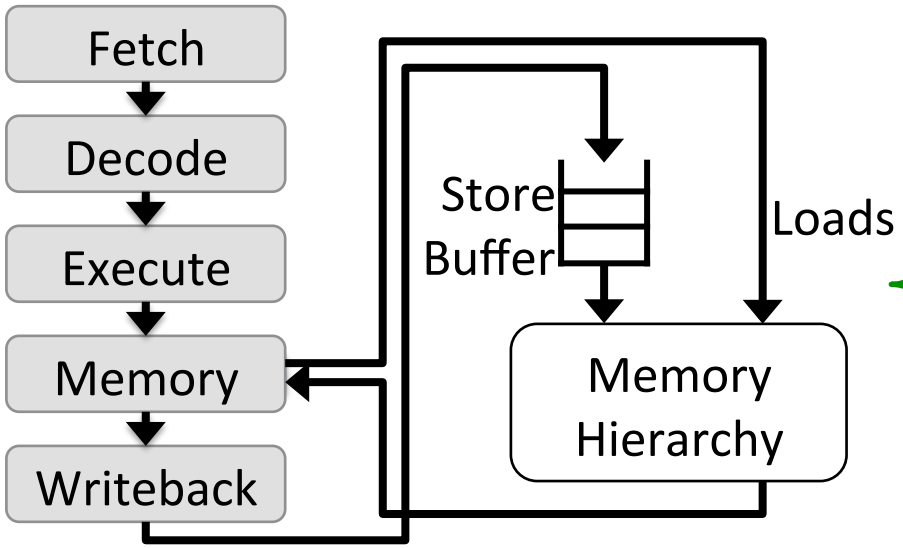
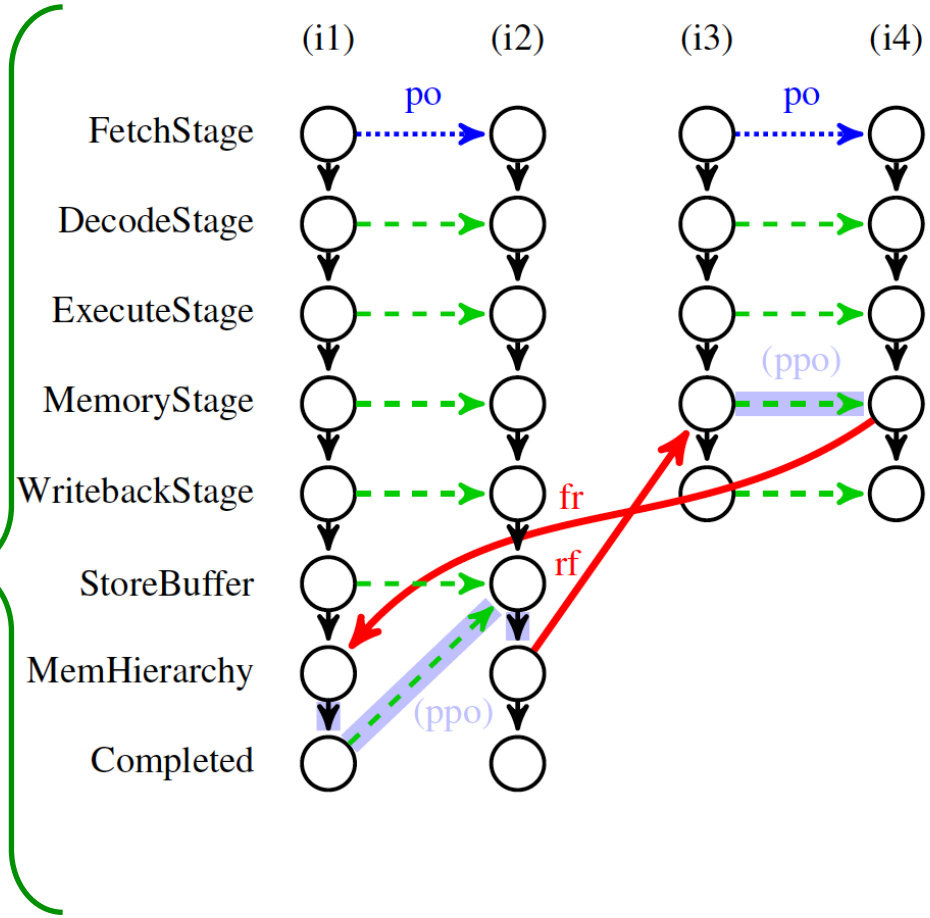
Microarchitectural Happens-Before Graphs

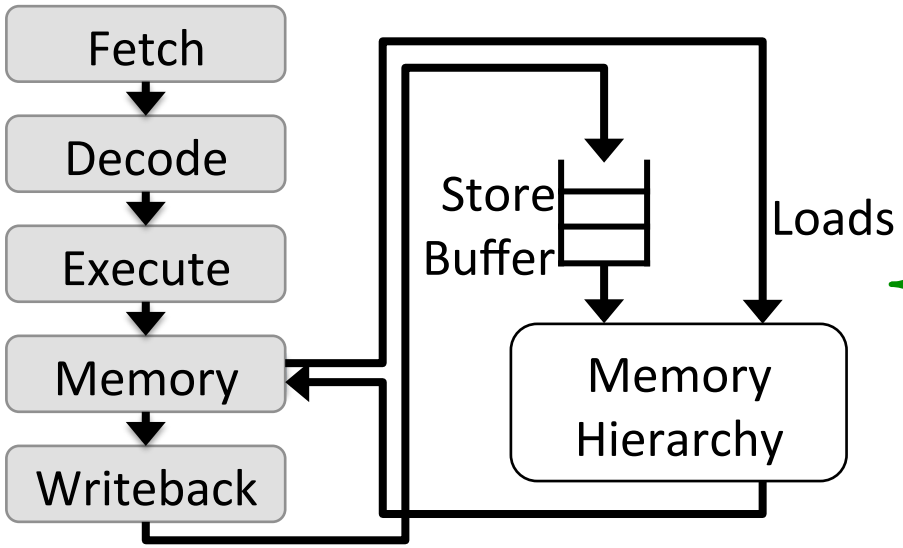


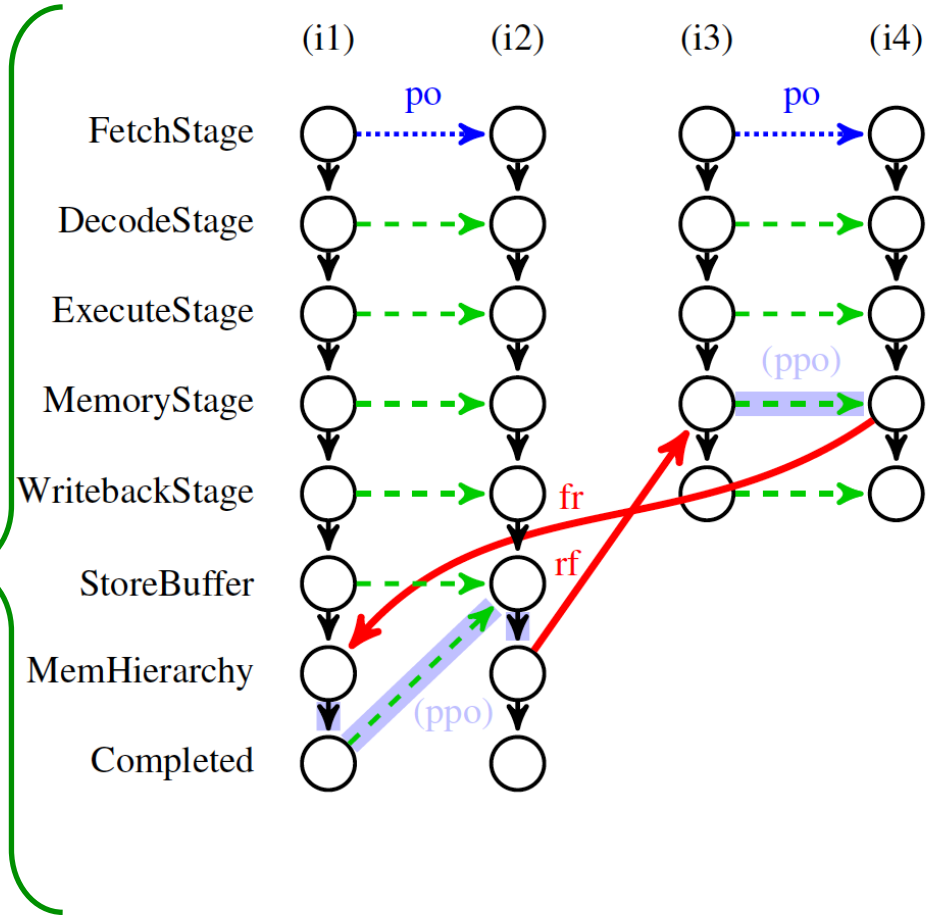


PipeCheck Overview

- Key idea: building ***microarchitectural*** happens-before graphs
- Inputs: arch. spec, µarch. spec, test suite
  - PPO Tests
  - Litmus Tests
- Approach: exhaustively enumerate and check all µhb graphs for cycles
  - Quick runtimes due to modest graph sizes

Microarchitectural Happens-Before Graphs





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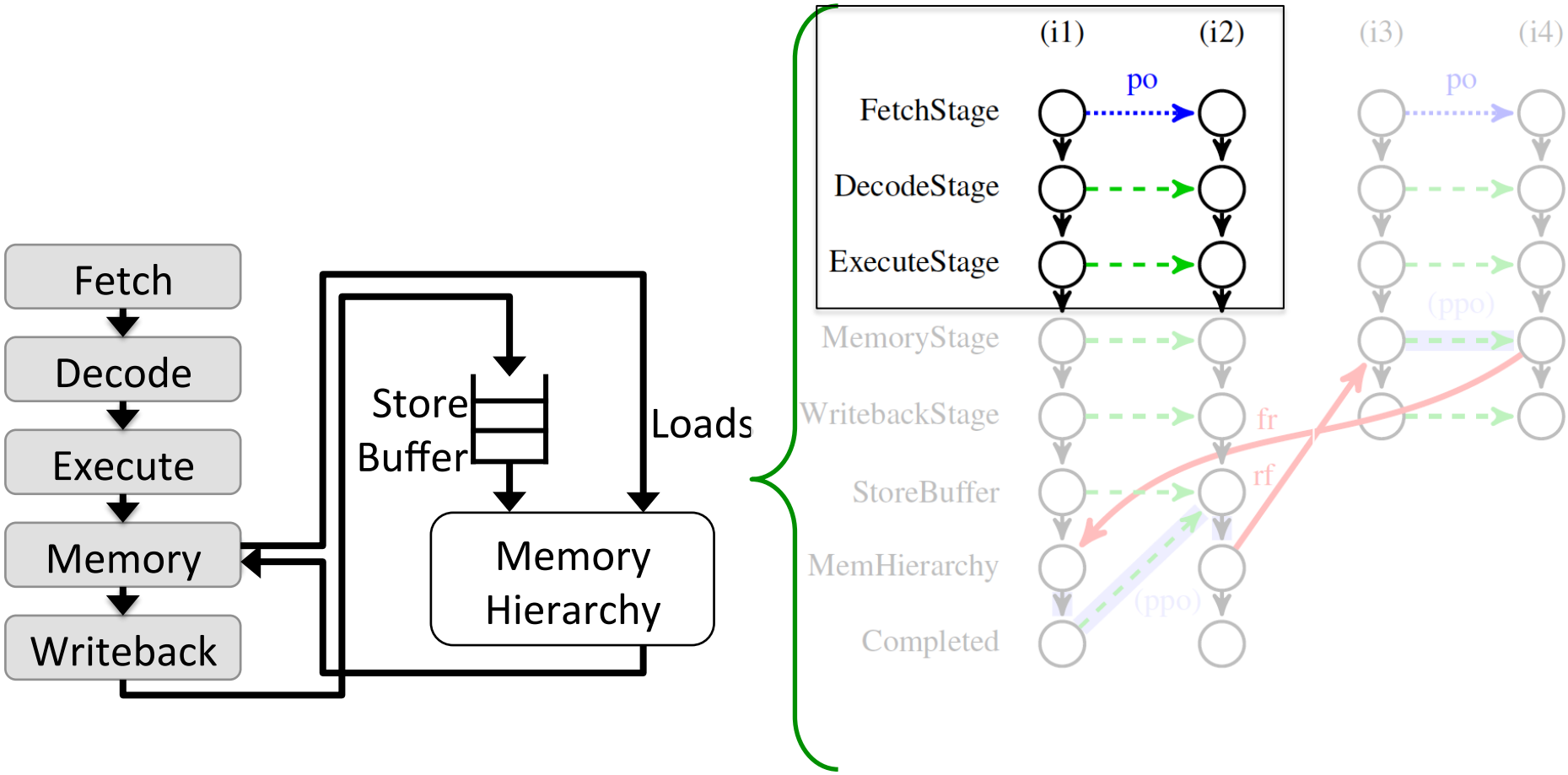
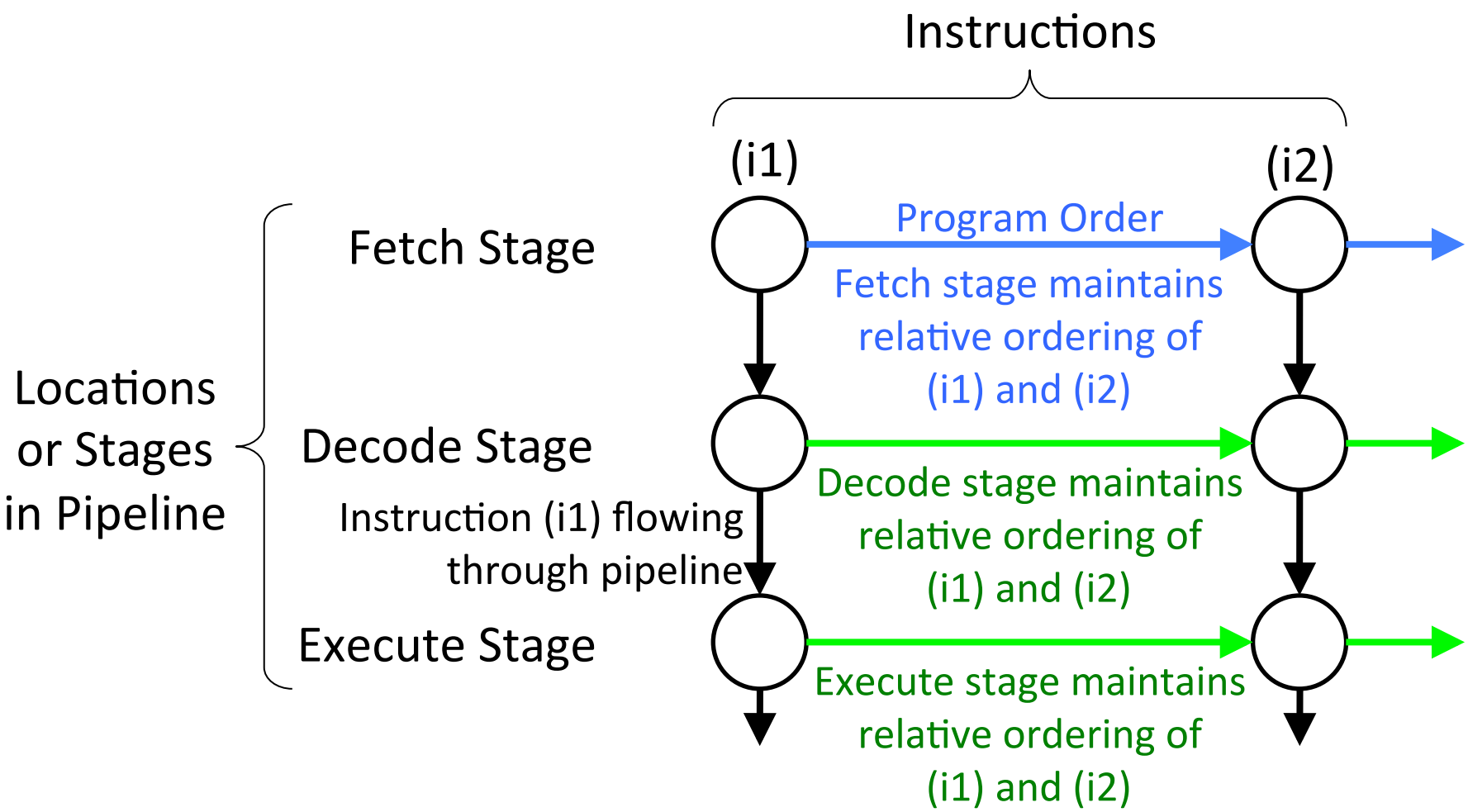


Lustig, Pellauer, Martonosi, “PipeCheck”

A Closer Look



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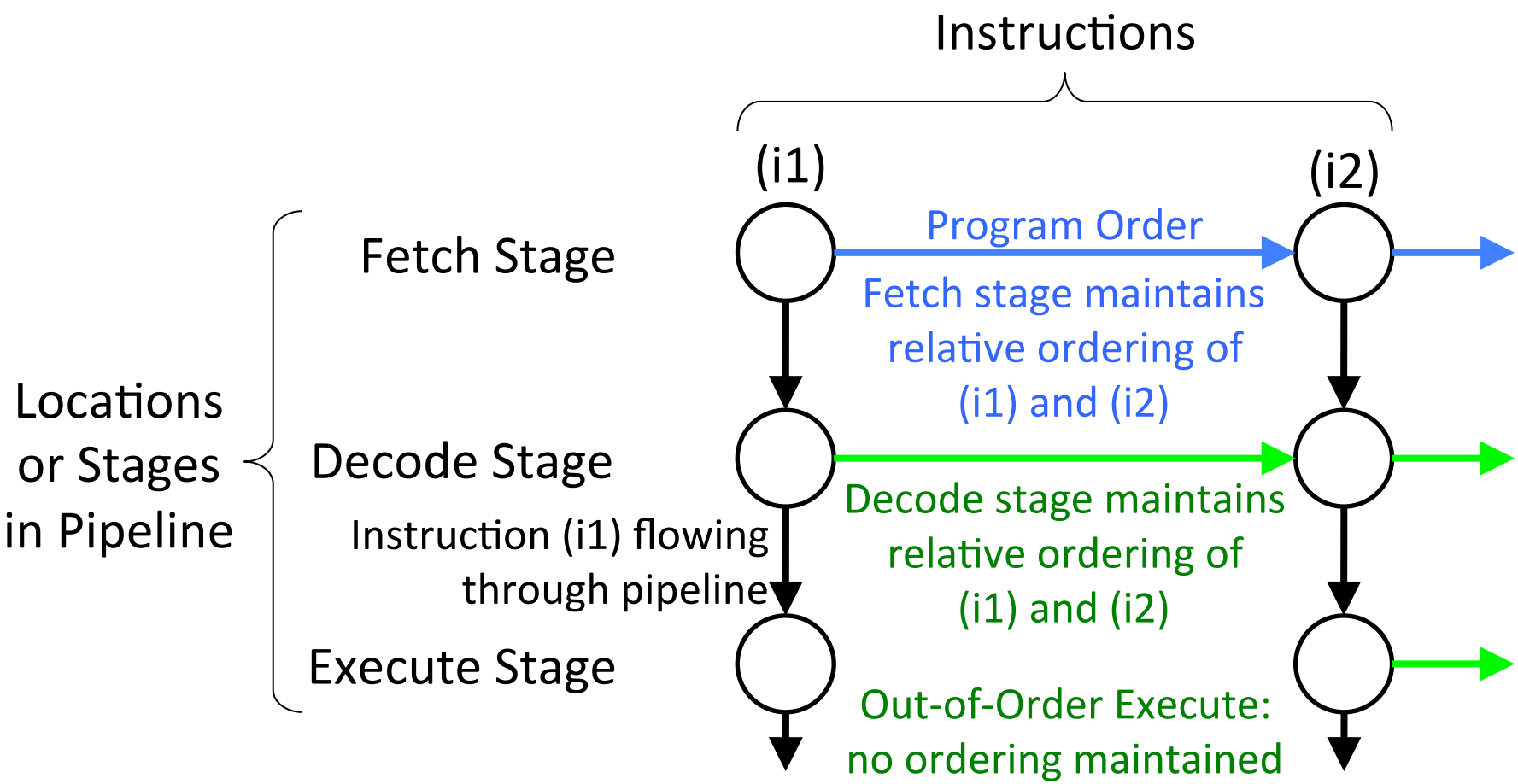

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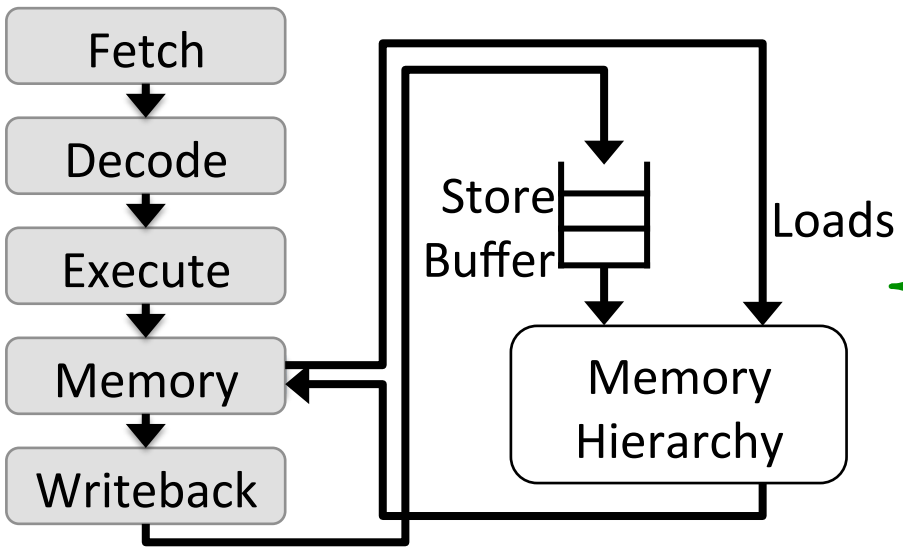
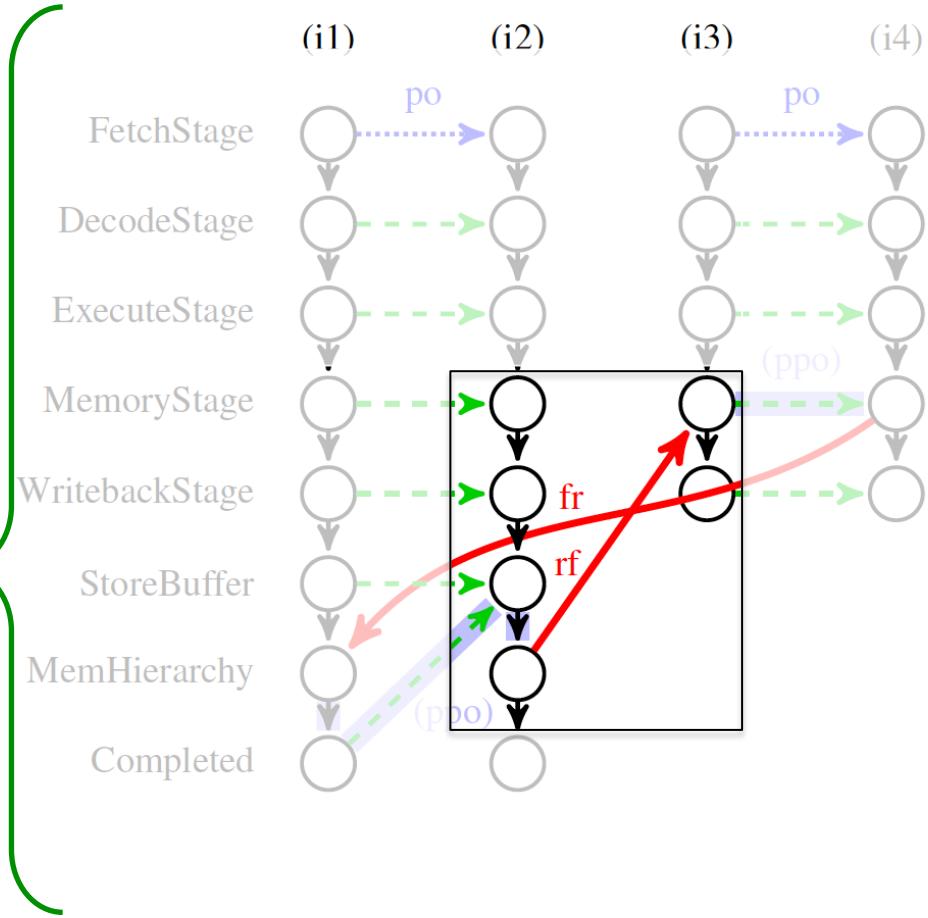
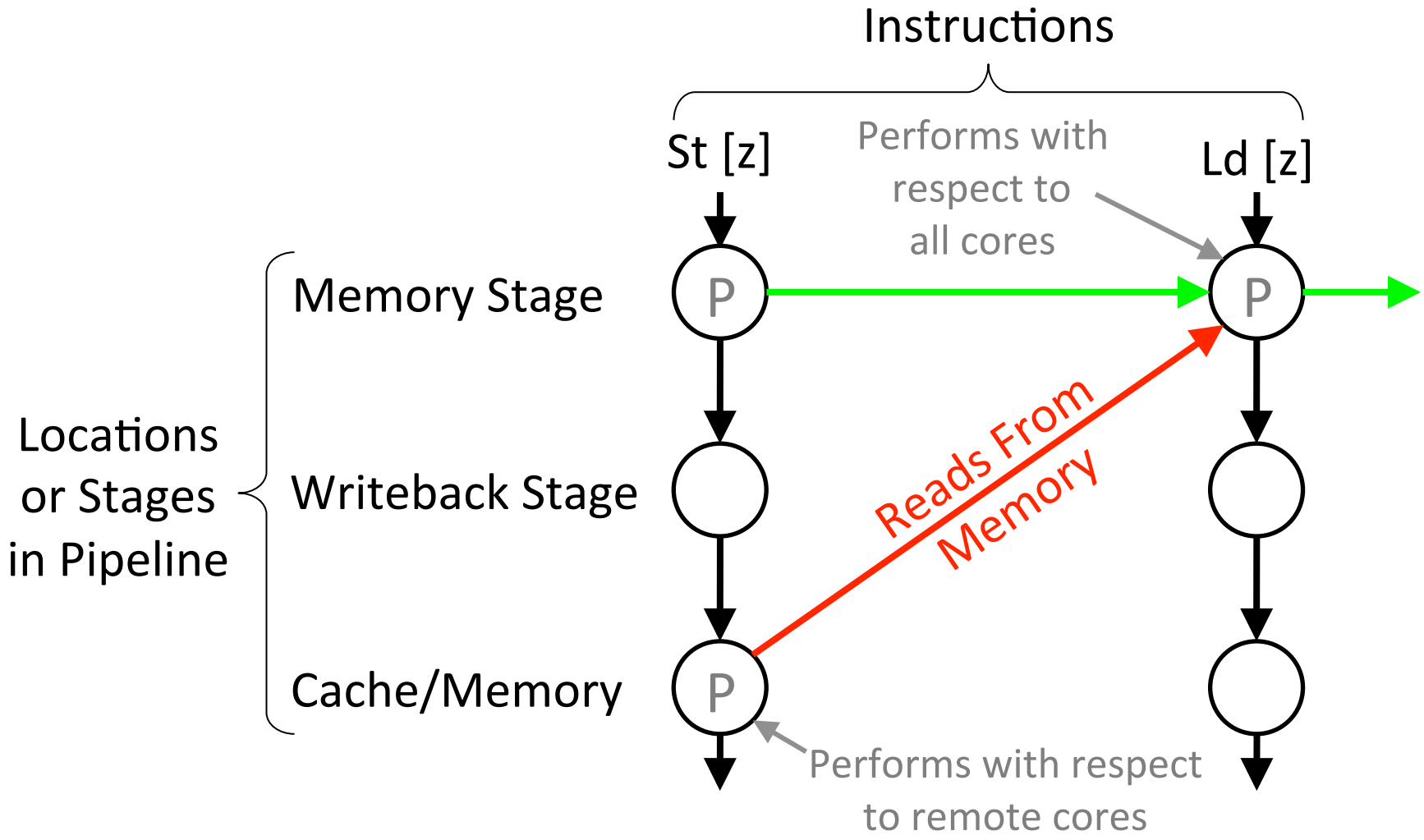




Reading From Particular Stages



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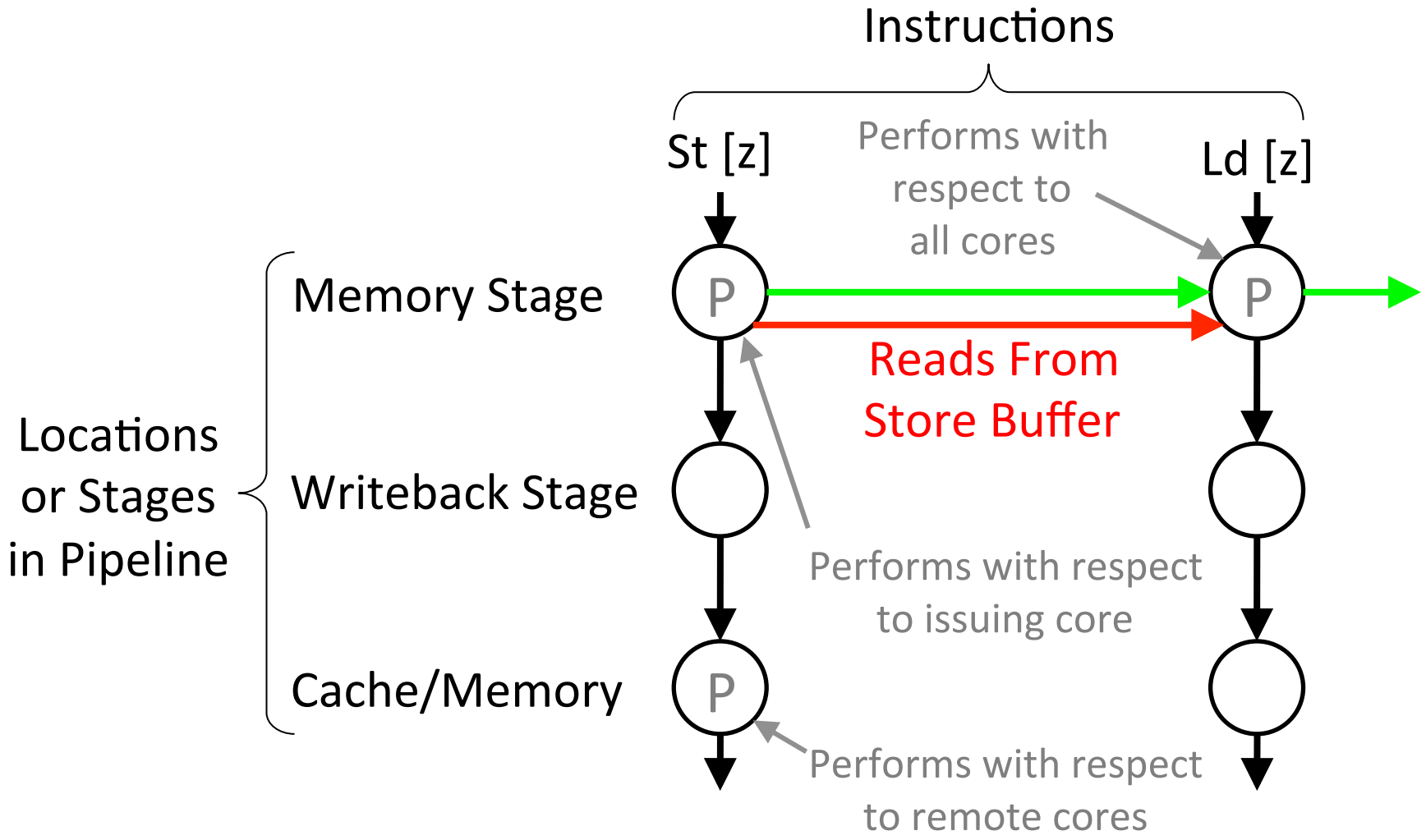

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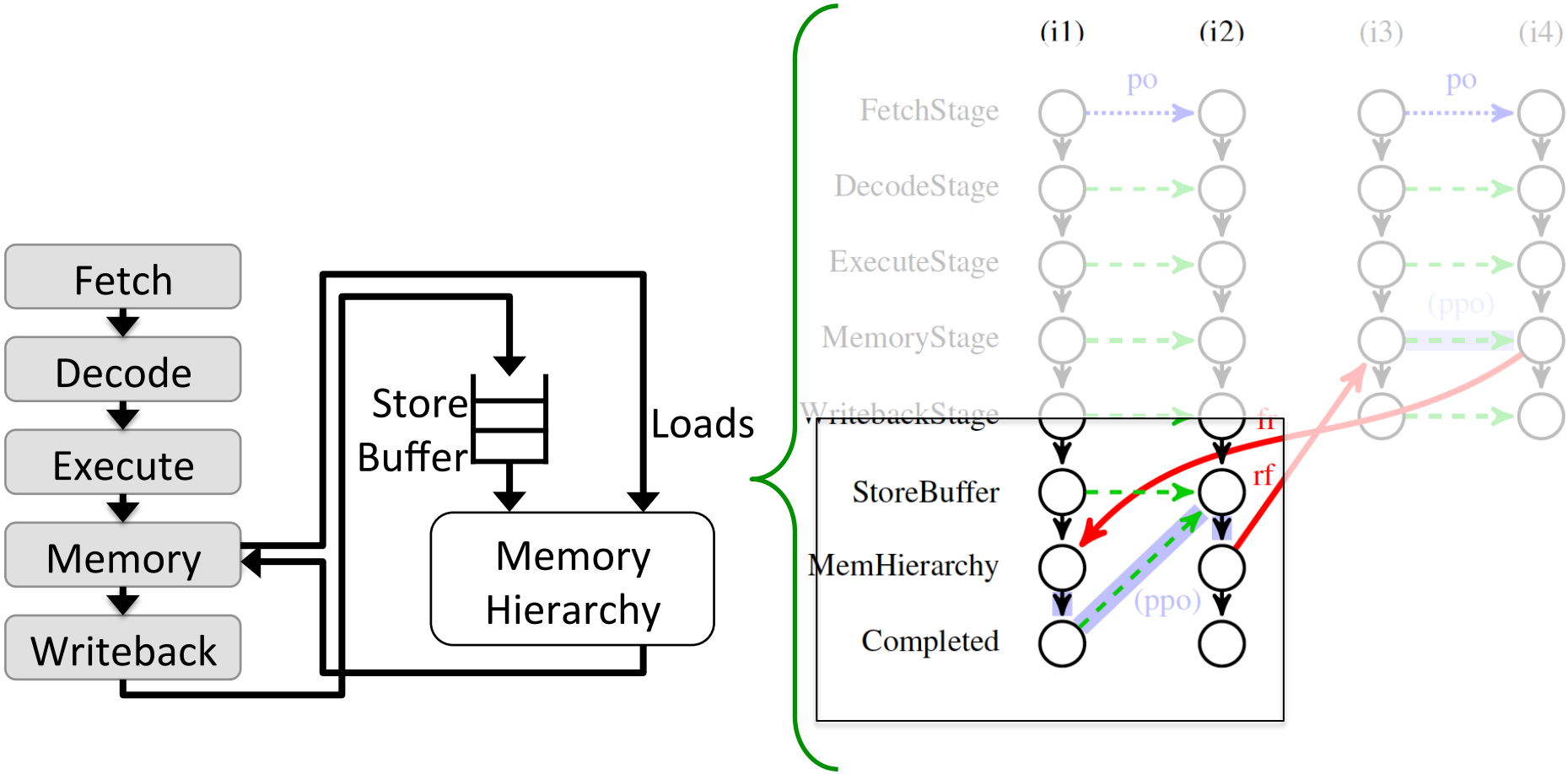
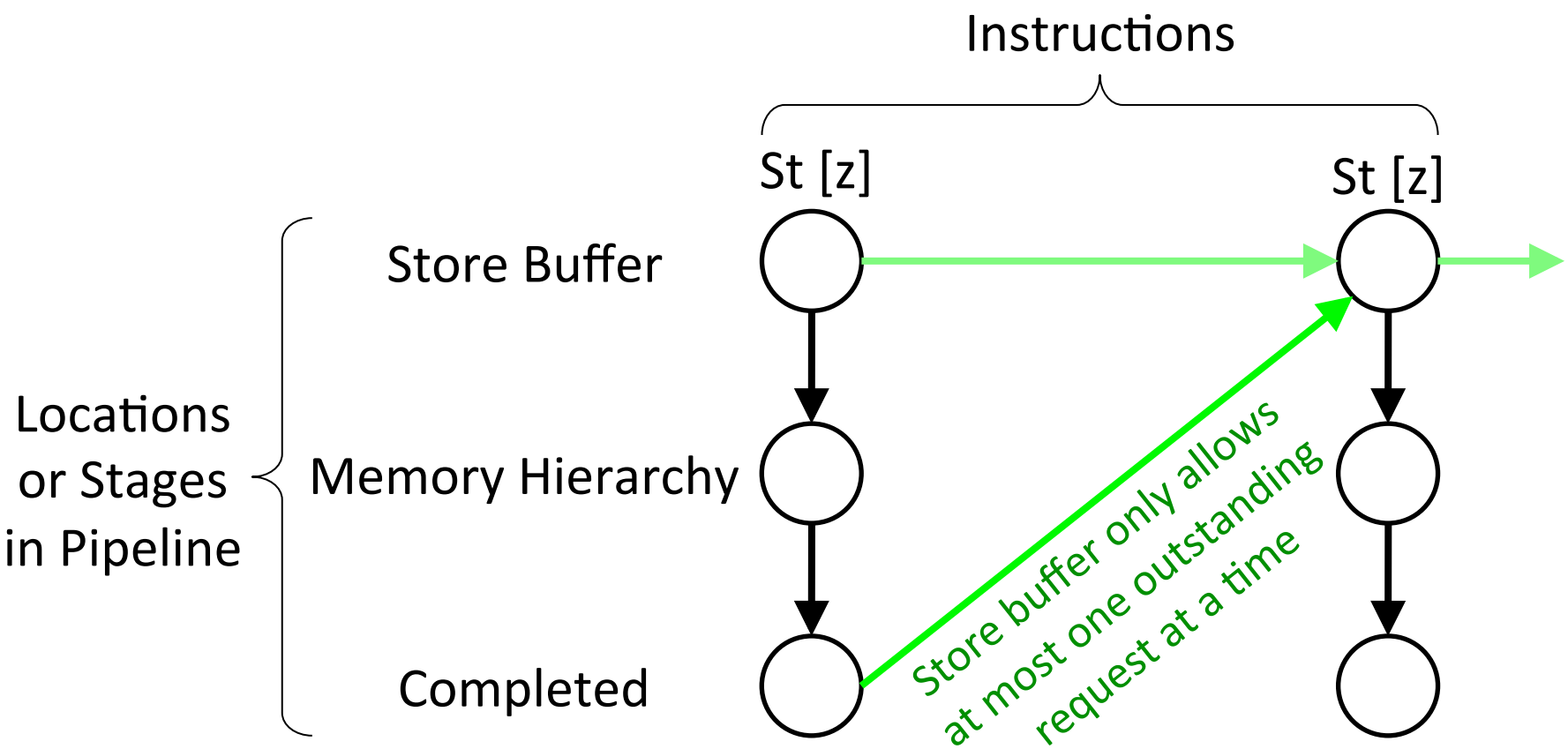


Verifying Preserved Program Order



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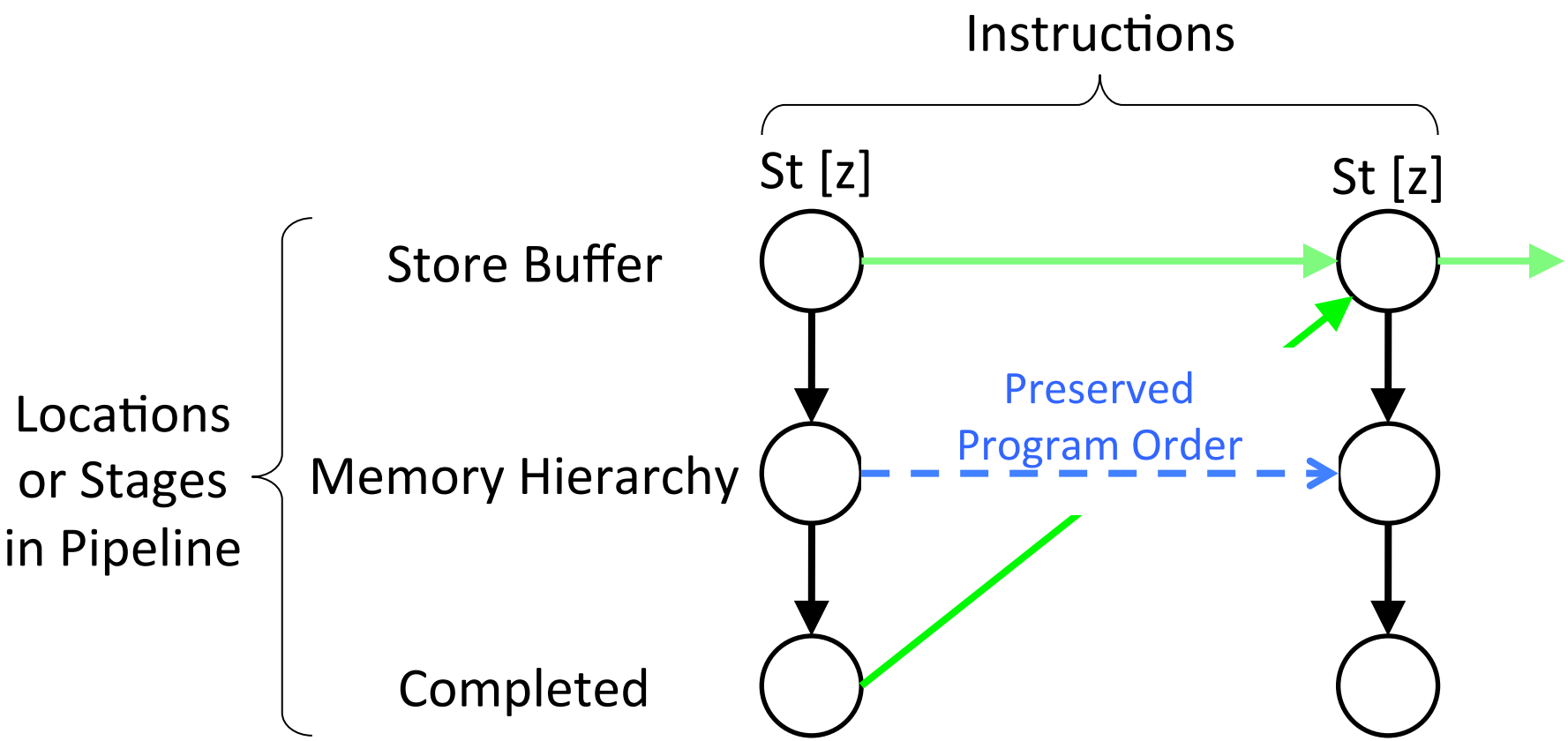

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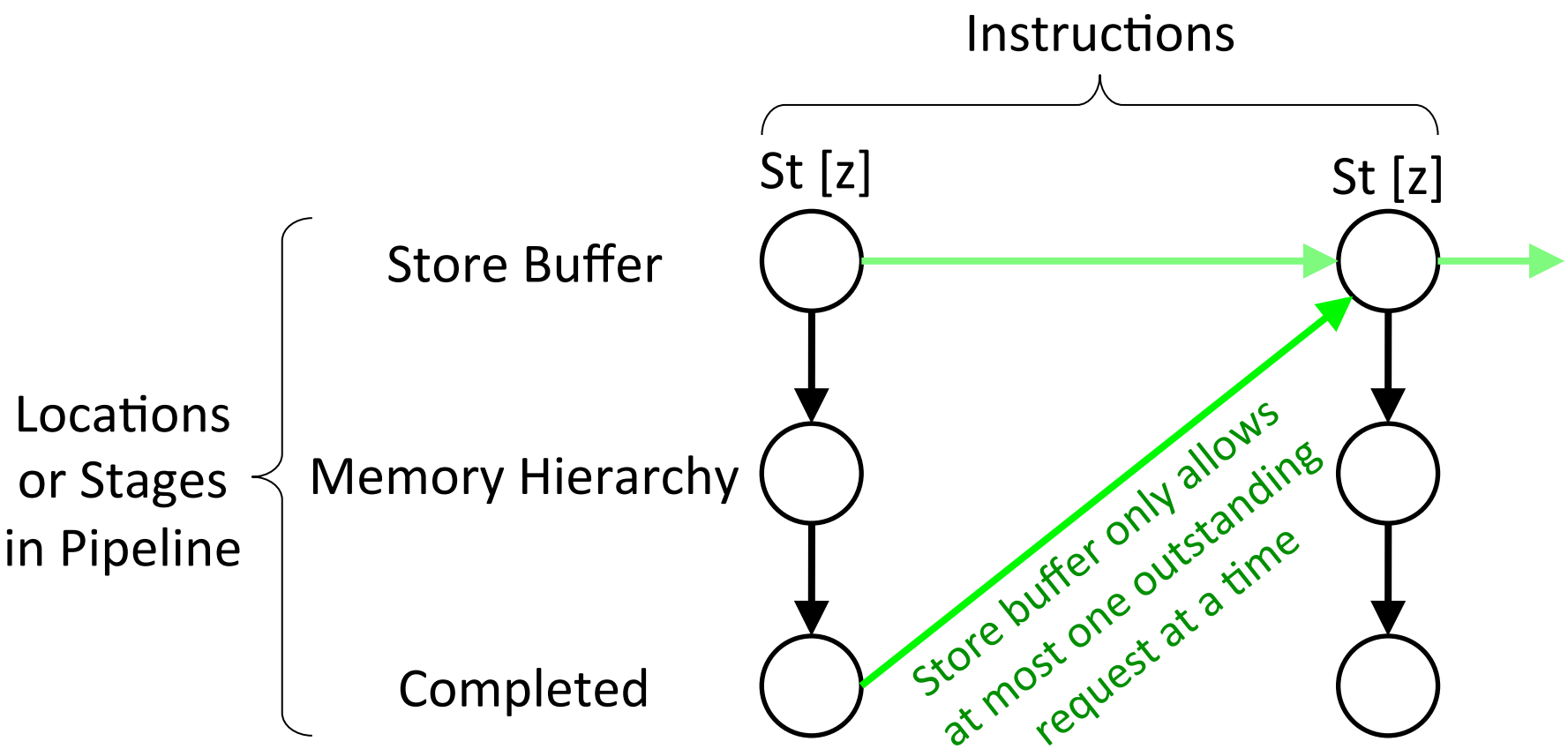

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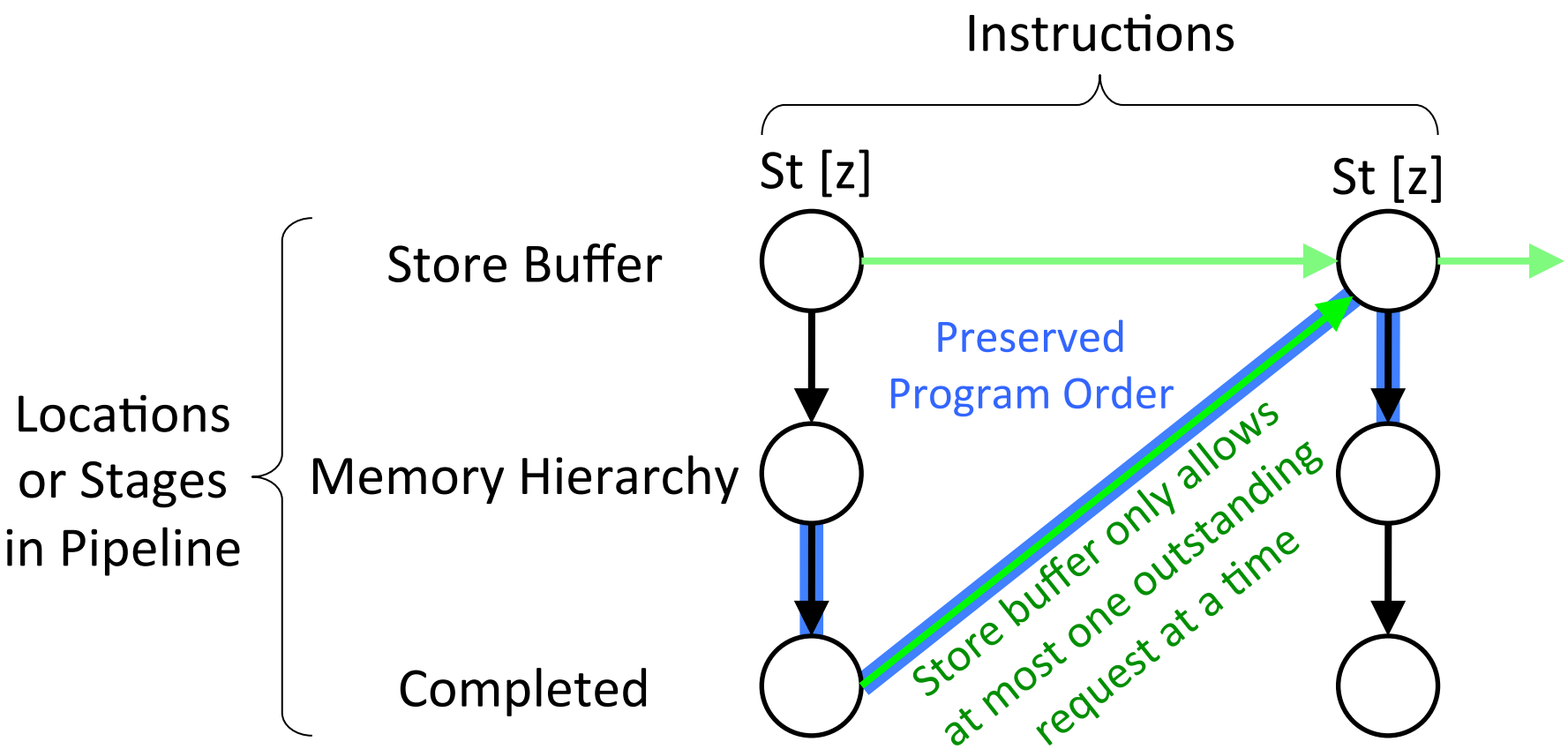
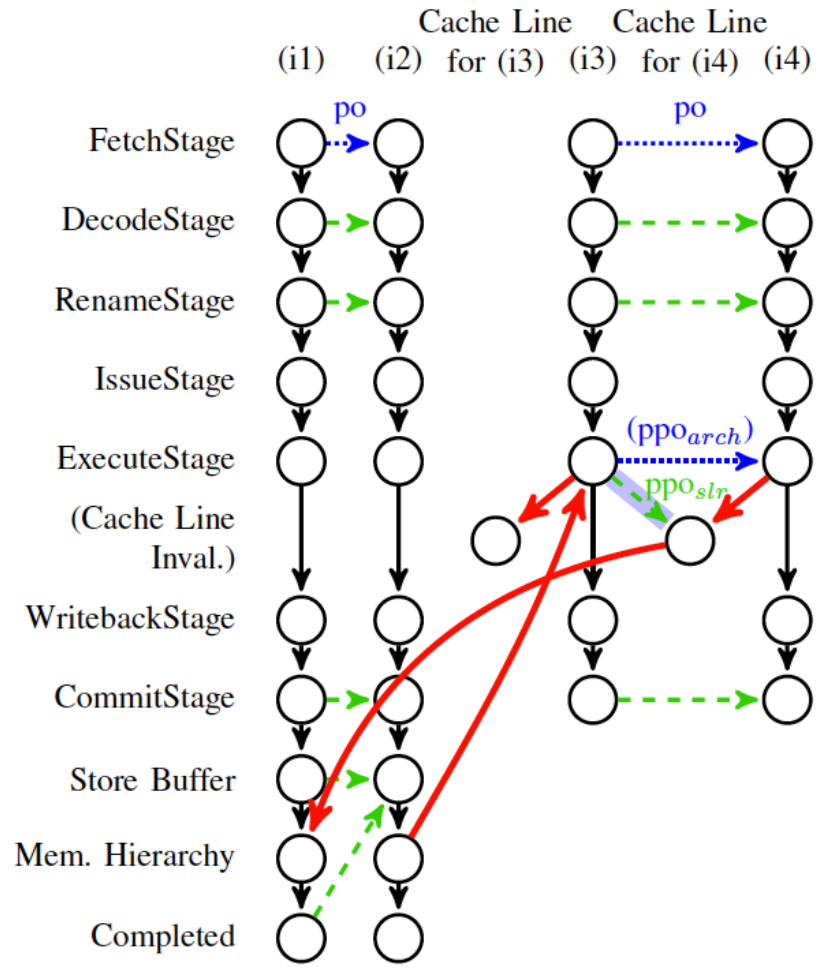


Microarchitectural Optimizations

PipeCheck also supports:

- OoO cores
- Speculative load reordering
- WeeFence
- Heterogeneity
- and so on



PipeCheck Software Implementation

- Tool written in Coq and extracted to OCaml
- PipeCheck software verifies each pipeline against suite of litmus tests and PPO tests
  - **Automatically** enumerate all possible executions (i.e. all possible graphs) for each pipeline model/ test pair

	Observable	Not Observable
Permitted	OK	OK (µarch stricter than necessary)
Forbidden	Pipeline bug!!	OK

PipeCheck Pipeline Models

- Pipelines modeled by specifying:
  - list of stages
  - list of possible paths through the pipeline
  - set of "non-local edges" (details in paper)
  - list of "performing locations"

Pipeline	Lines of Code
Classic 5-Stage Pipeline without Store Buffer	37
Classic 5-Stage Pipeline with Store Buffer	62
gem5 O3 CPU Model	106
OpenSPARC T2	115

PipeCheck Litmus Test Results

Litmus Test	Expected	5-Stage w/o St. Buf	5-Stage w/ St. Buf	gem5 O3	OpenSPARC T2
iwp2.1/amd1	Forbid	✔	✔	Observed[2]	✔
iwp2.2/amd2	Forbid	✔	✔	✔	✔
iwp2.3a/amd4	Permit	Not obs.[1]	✔	✔	✔
iwp2.3b	Permit	✔	✔	✔	✔
iwp2.4/amd9	Permit	Not obs.[1]	✔	✔	✔
iwp2.5/amd8	Forbid	✔	✔	Observed[2]	✔
iwp2.6	Forbid	✔	✔	✔	✔
amd3	Permit	Not obs.[1]	✔	✔	✔
amd6	Forbid	✔	✔	Observed[2]	✔
n1	Permit	Not obs.[1]	✔	✔	✔
n2	Forbid	✔	✔	Observed[2]	✔
n4	Forbid	✔	✔	✔	✔
n5	Forbid	✔	✔	✔	✔
n6	Permit	✔	✔	✔	✔
n7	Permit	Not obs.[1]	✔	✔	✔
rwc-unfenced	Permit	Not obs.[1]	✔	✔	✔

1. Permitted results not observable: pipeline stronger than necessary

2. Forbidden results observable: found bugs in pipeline!

PipeCheck Litmus Test Results

Litmus Test	Expected	5-Stage w/o St. Buf	5-Stage w/ St. Buf	gem5 O3	OpenSPARC T2
iwp2.1/amd1	Forbid	✔	✔	**Observed[2]**	✔
iwp2.2/amd2	Forbid	✔	✔	✔	✔
iwp2.3a/amd4	Permit	**Not obs.[1]**	✔	✔	✔
iwp2.3b	Permit	✔	✔	✔	✔
iwp2.4/amd9	Permit	**Not obs.[1]**	✔	✔	✔
iwp2.5/amd8	Forbid	✔	✔	**Observed[2]**	✔
iwp2.6	Forbid	✔	✔	✔	✔
amd3	Permit	**Not obs.[1]**	✔	✔	✔
amd6	Forbid	✔	✔	**Observed[2]**	✔
n1	Permit	**Not obs.[1]**	✔	✔	✔
n2	Forbid	✔	✔	**Observed[2]**	✔
n4	Forbid	✔	✔	✔	✔
n5	Forbid	✔	✔	✔	✔
n6	Permit	✔	✔	✔	✔
n7	Permit	**Not obs.[1]**	✔	✔	✔
rwc-unfenced	Permit	**Not obs.[1]**	✔	✔	✔

1. Permitted results not observable: pipeline stronger than necessary

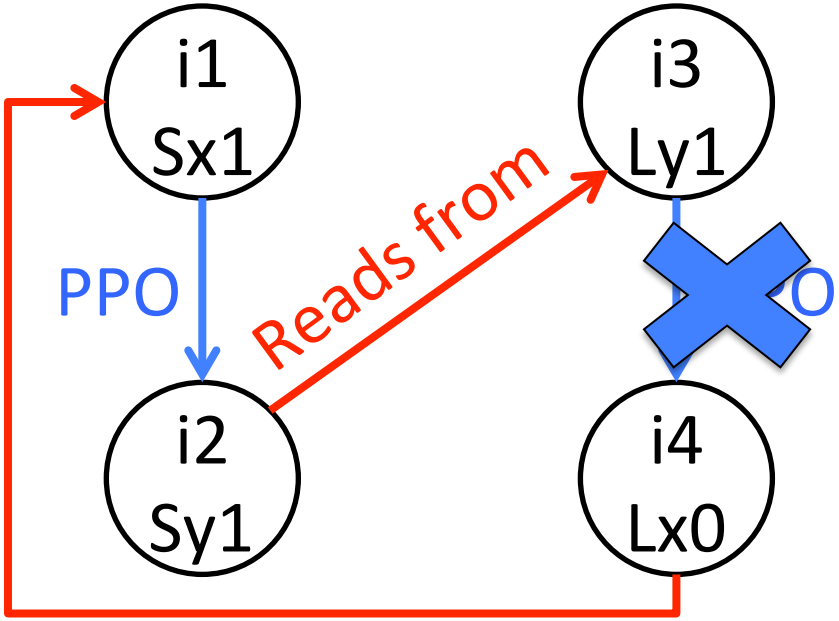
2. Forbidden results observable: found bugs in pipeline!

Interpreting Bugs Found By PipeCheck

Initially: [x]=[y]=0

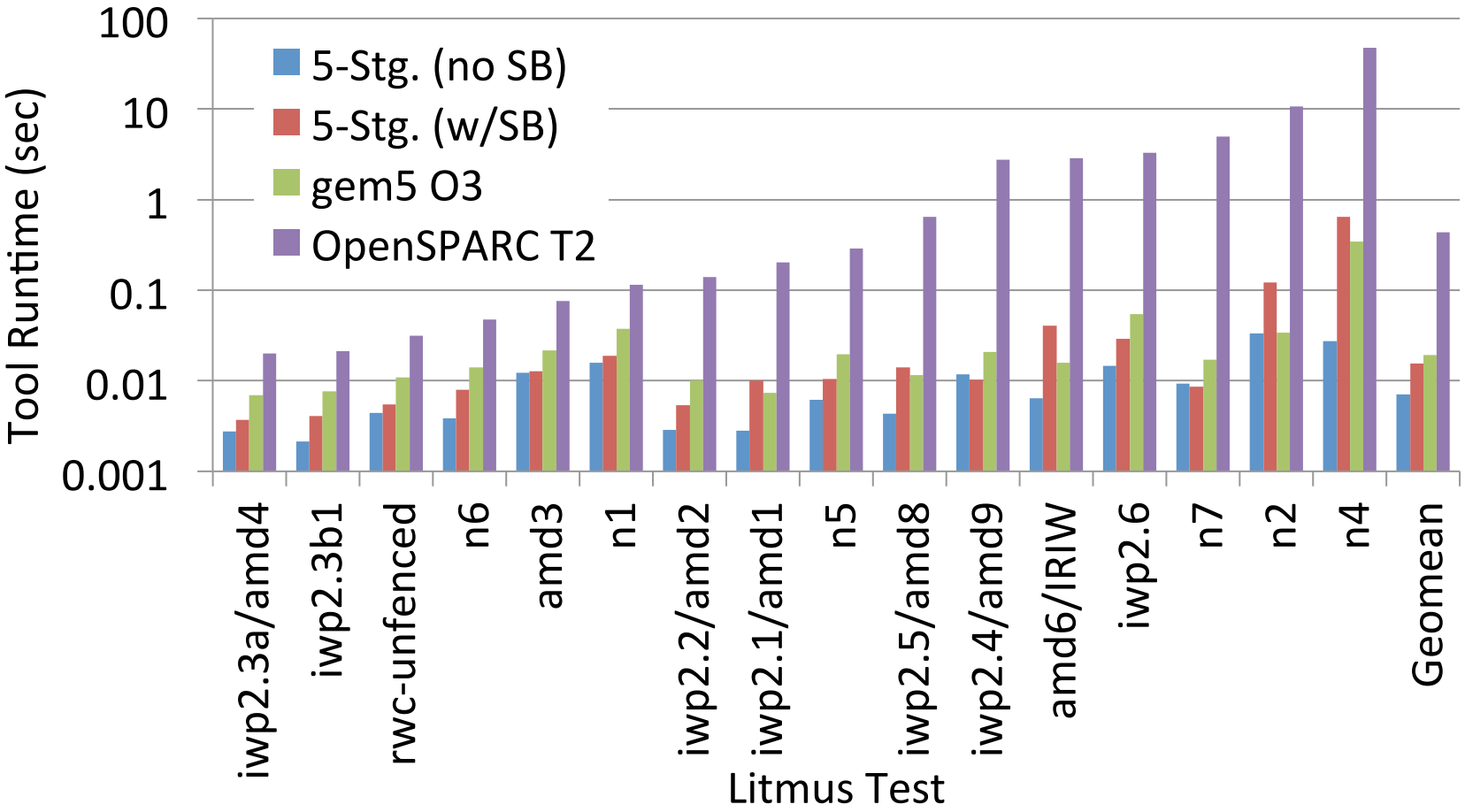
Core 0	Core 1
(i1) st [x], 1 (i2) st [y], 1	(i3) ld [y] → r1 (i4) ld [x] → r2

TSO: **Forbid**: r1=1, r2=0



Reads from earlier store
("from-reads")

PipeCheck Verification Time



PipeCheck Conclusions

- PipeCheck verifies the memory model ***implementation*** against vendor spec
  - Automated tool with practical runtimes
- Opens door to systematized checking of heterogeneous mix of memory referencers: CPU, GPU, accelerators, or anything else

PipeCheck: Specifying and Verifying Microarchitectural Enforcement of Memory Consistency Models

Daniel Lustig[+], Michael Pellauer[*],
and Margaret Martonosi[+]

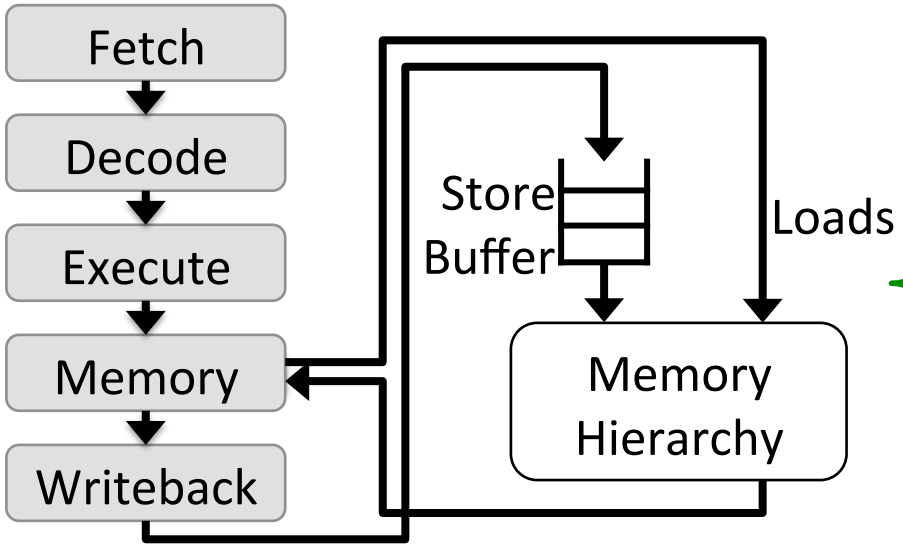
[+]Princeton University [*]Intel VSSAD

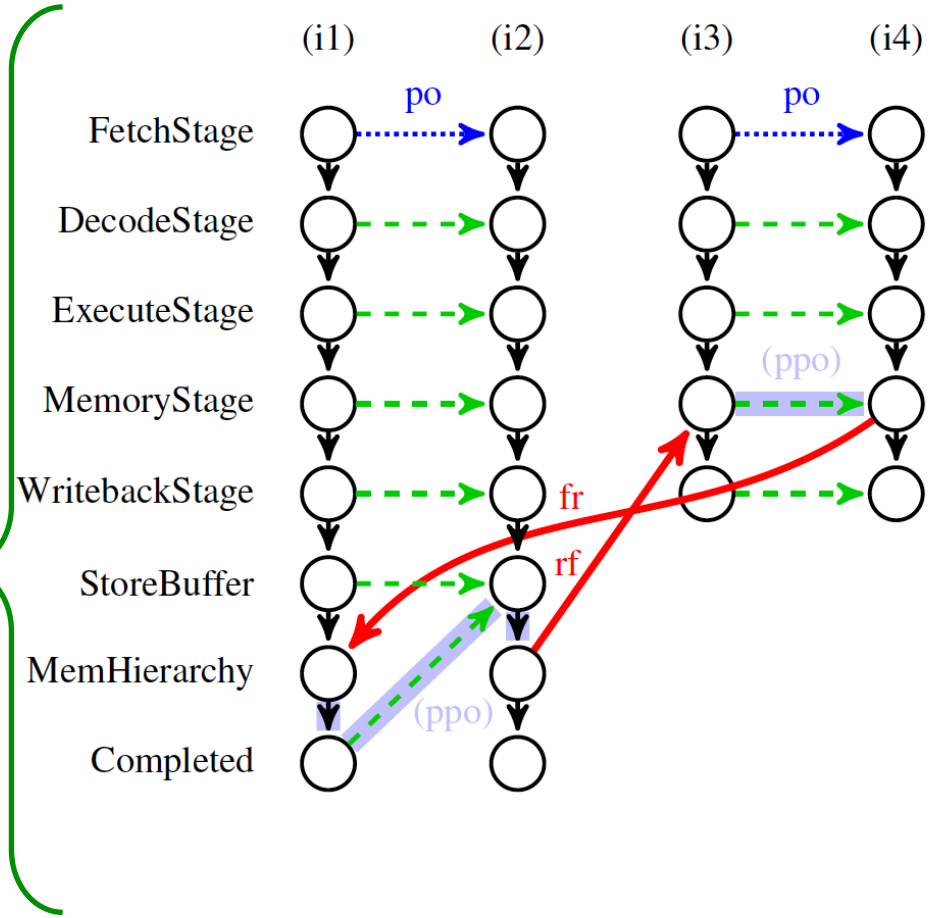
MICRO 47

PRINCETON

PipeCheck Overview

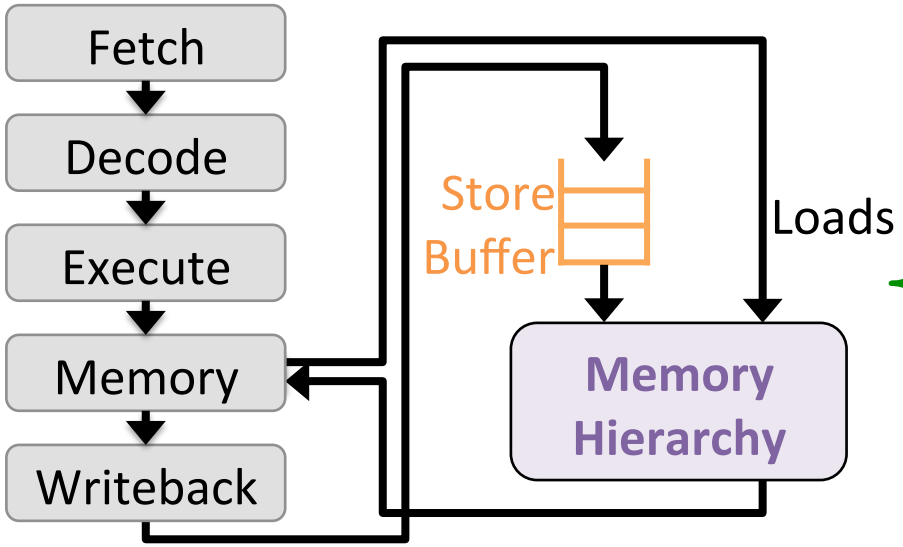
- **Microarchitectural equivalent** of "happens-before" graphs

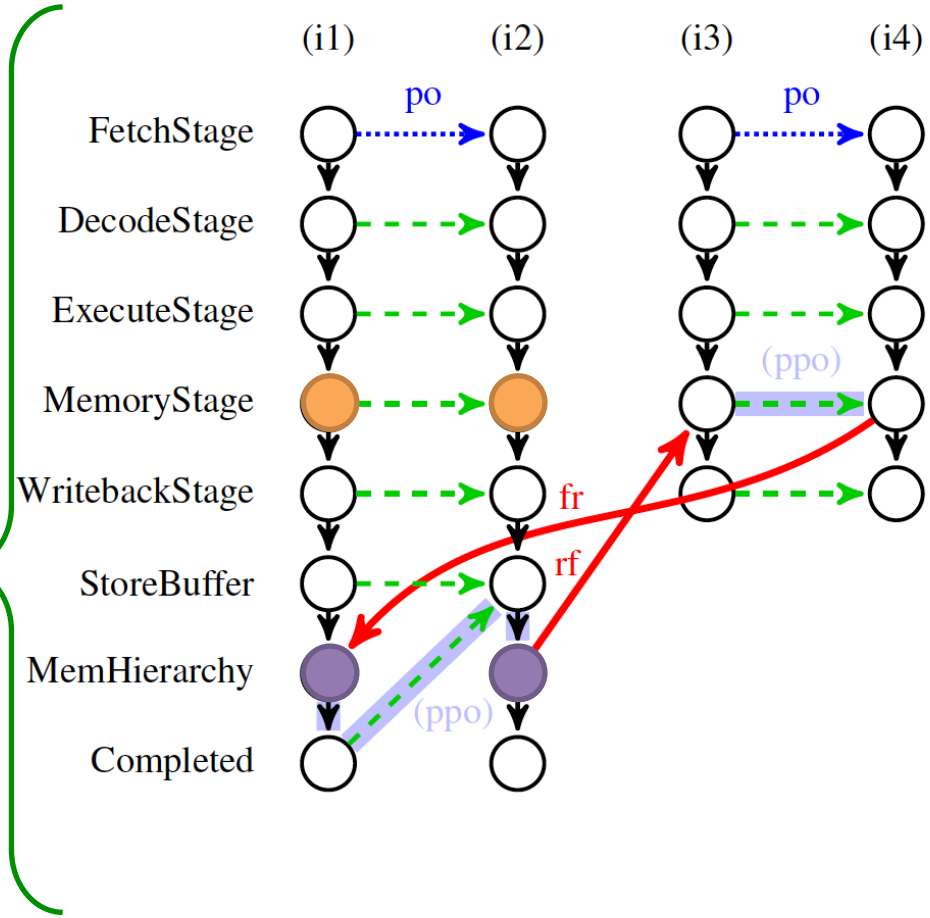




PipeCheck Overview

- **Microarchitectural equivalent** of "happens-before" graphs



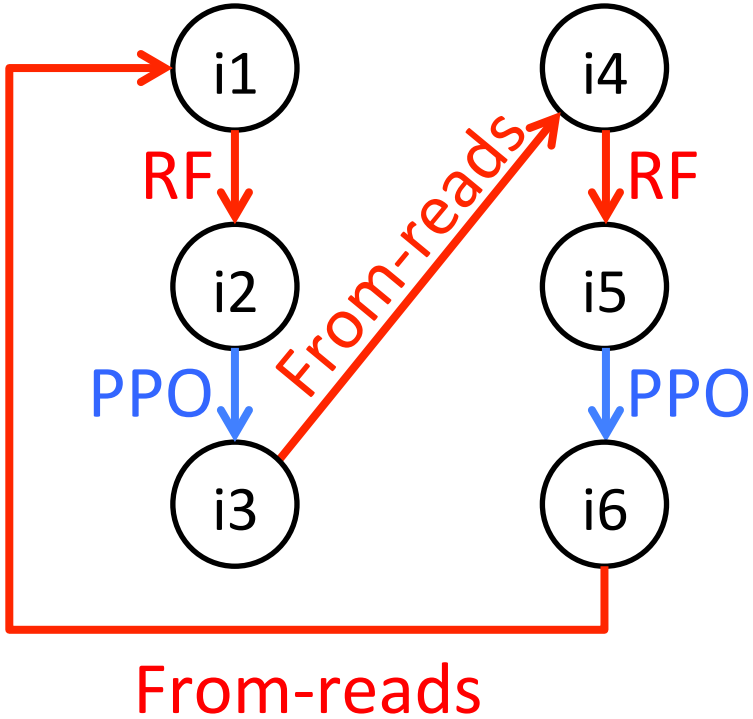


Non-SC Models: Not Every Cycle Forbids a Proposed Outcome

Litmus test iwp2.4/amd9:

Core 0	Core 1
(i1) st [x], 1	(i4) st [y], 1
(i2) ld [x] → r1	(i5) ld [y] → r3
(i3) ld [y] → r2	(i6) ld [x] → r4

Outcome: r1=r3=1, r2=r4=0
Observed on Hardware



[Intel, AMD, Alglave, FMSD '12, Owens et al., TPHOLs '09]

More Complexity = Special Cases

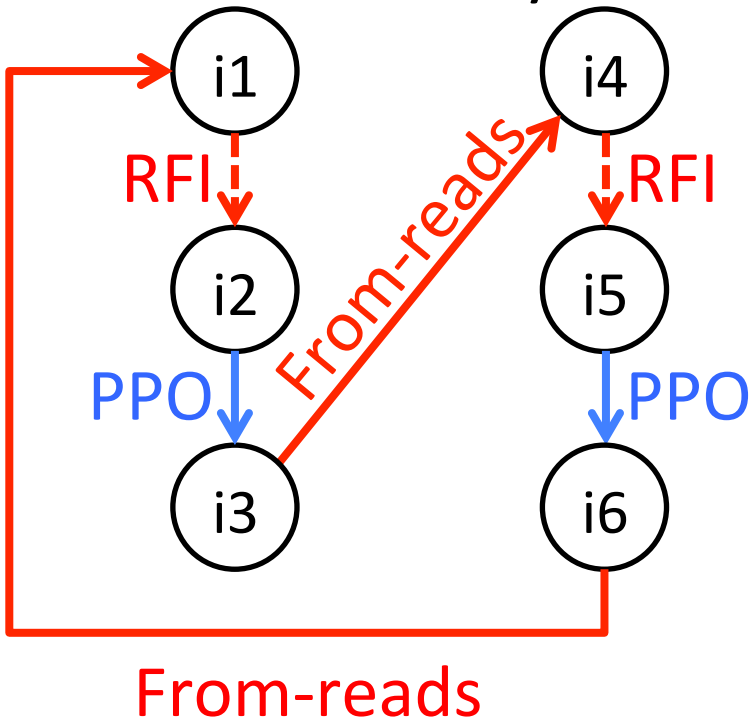
Litmus test iwp2.4/amd9:

Core 0	Core 1
(i1) st [x], 1	(i4) st [y], 1
(i2) ld [x] → r1	(i5) ld [y] → r3
(i3) ld [y] → r2	(i6) ld [x] → r4

TSO: **Allow** r1=r3=1, r2=r4=0

To fully capture all model details, need either complex nodes or complex **edges**

Architectural Analysis:



[Alglave, FMSD '12, Owens et al., TPHOLs '09]

More Complexity = Special Cases

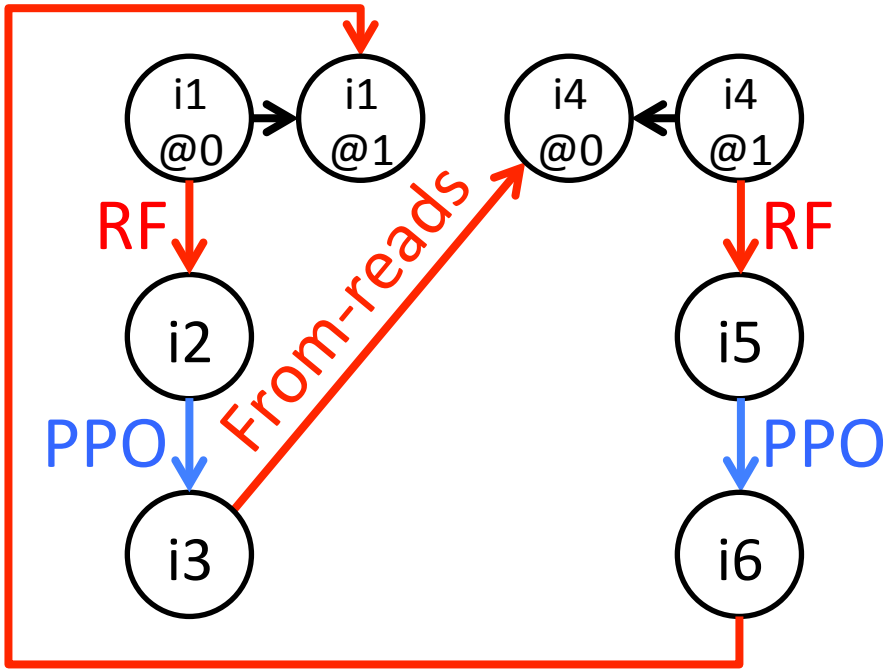
Litmus test iwp2.4/amd9:

Core 0	Core 1
(i1) st [x], 1 (i2) ld [x] → r1 (i3) ld [y] → r2	(i4) st [y], 1 (i5) ld [y] → r3 (i6) ld [x] → r4

TSO: **Allow** r1=r3=1, r2=r4=0

To fully capture all model details, need either complex **nodes** or complex edges

Architectural Analysis:



From-reads

What is the Definition of "Correct"?

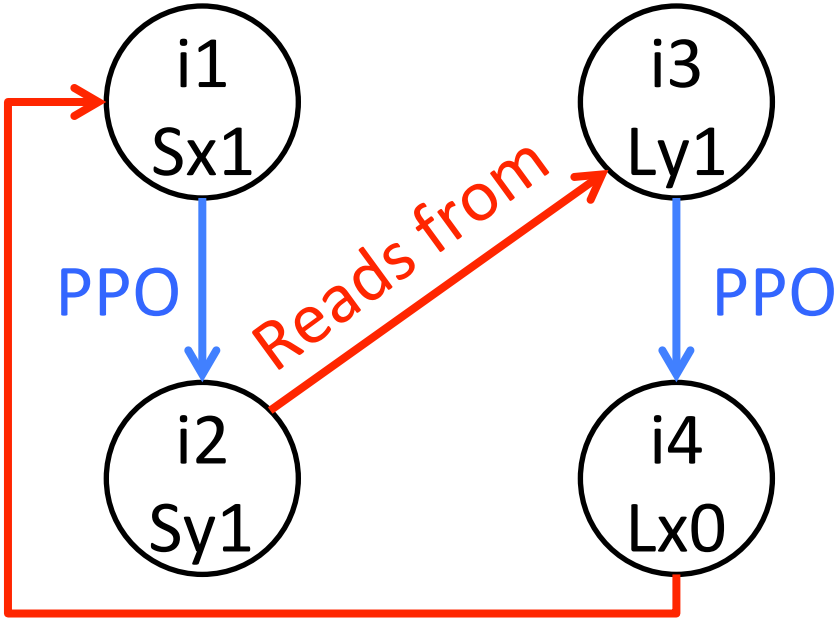
Litmus test mp:

Initially: [x]=[y]=0

Core 0	Core 1
(i1) st [x], 1 (i2) st [y], 1	(i3) ld [y] → r1 (i4) ld [x] → r2

TSO: **Forbid**: r1=1, r2=0

Cycle implies forbidden execution: instruction can't happen before itself

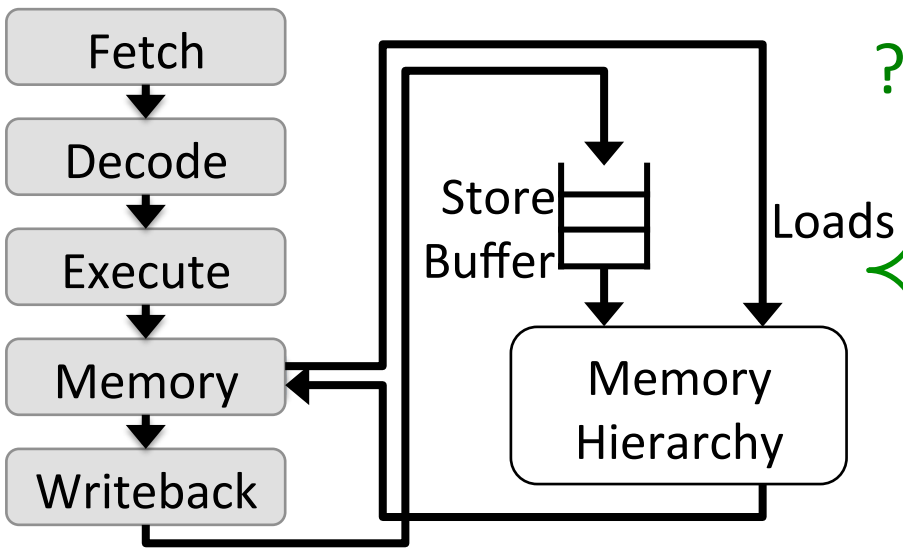


Reads from earlier store ("from-reads")

[MD, Alglave, FMSD '12, Owens et al., TPHOLs '09]

Insights from Microarchitecture

- How is the consistency model **enforced at the microarchitecture level**?



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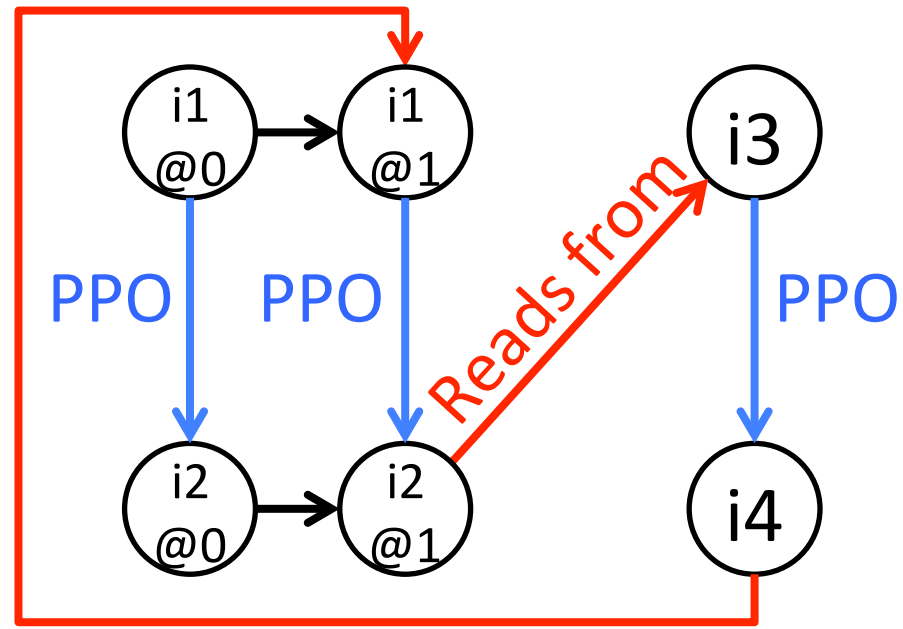

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Insights from Microarchitecture

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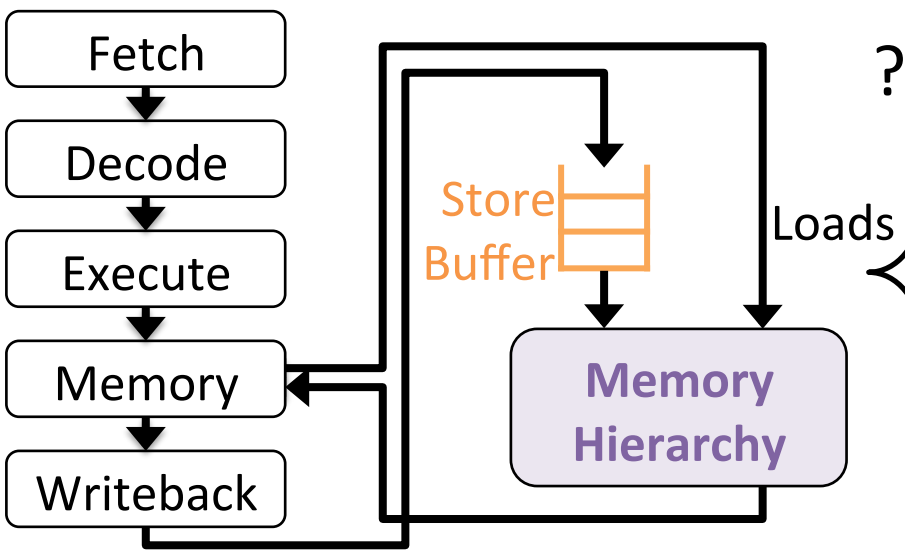

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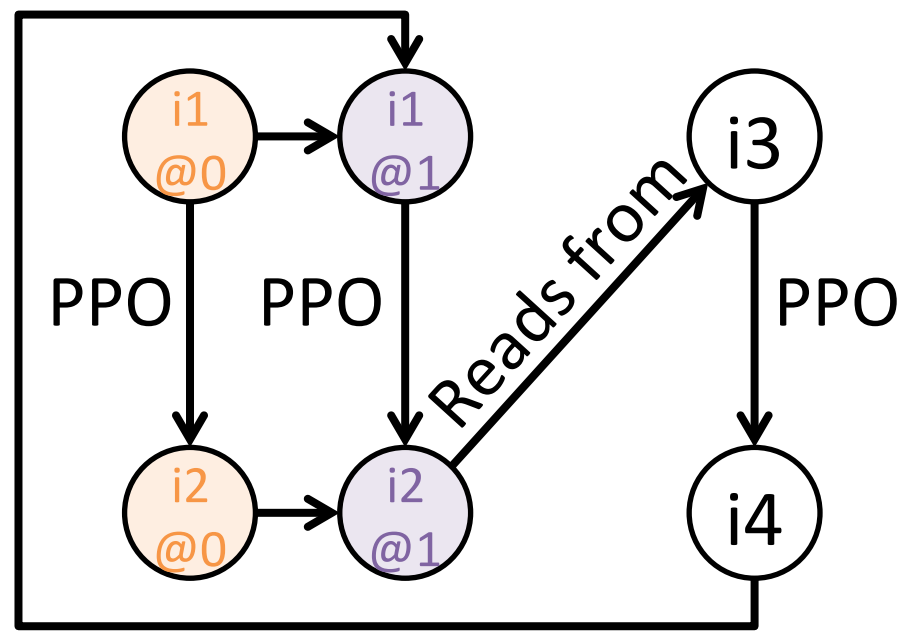

Reads from earlier ("from-reads")

Insights from Microarchitecture

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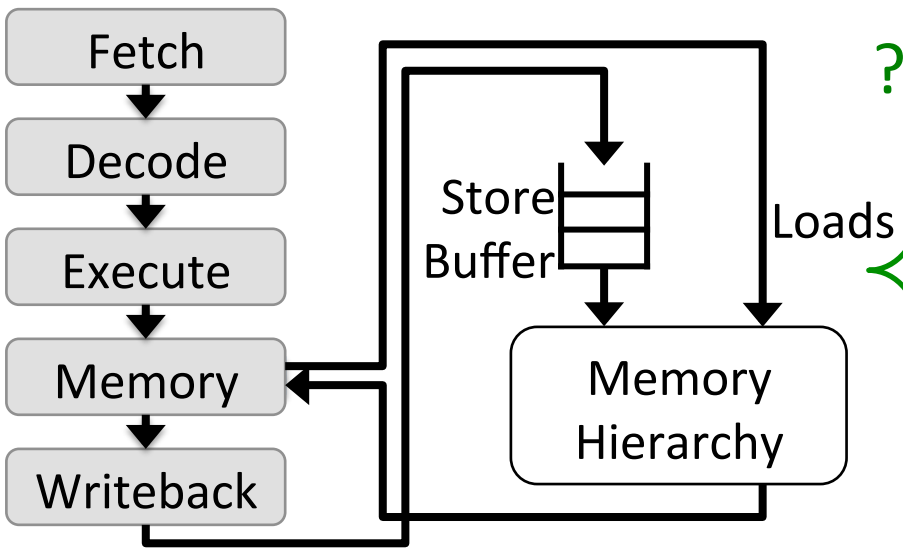

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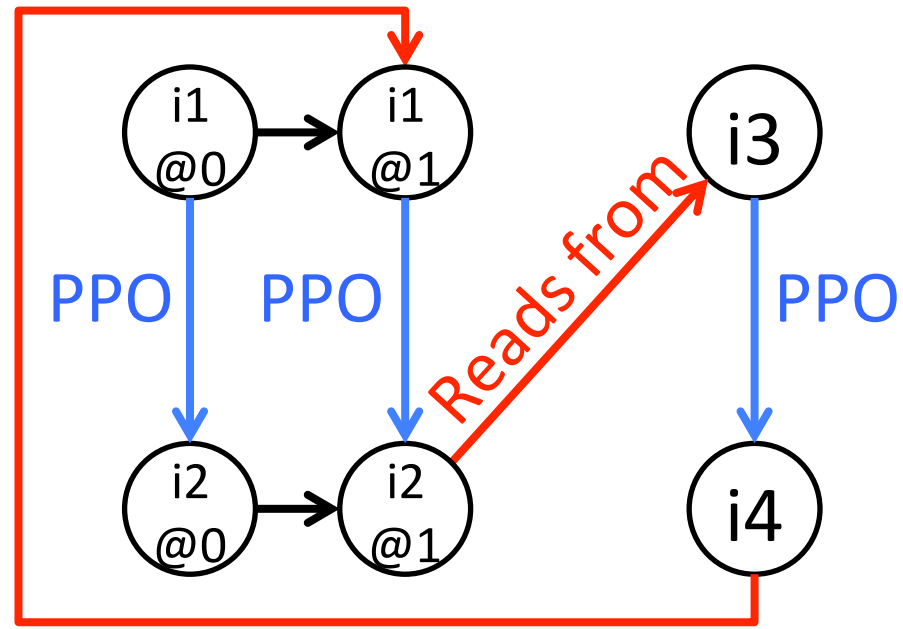
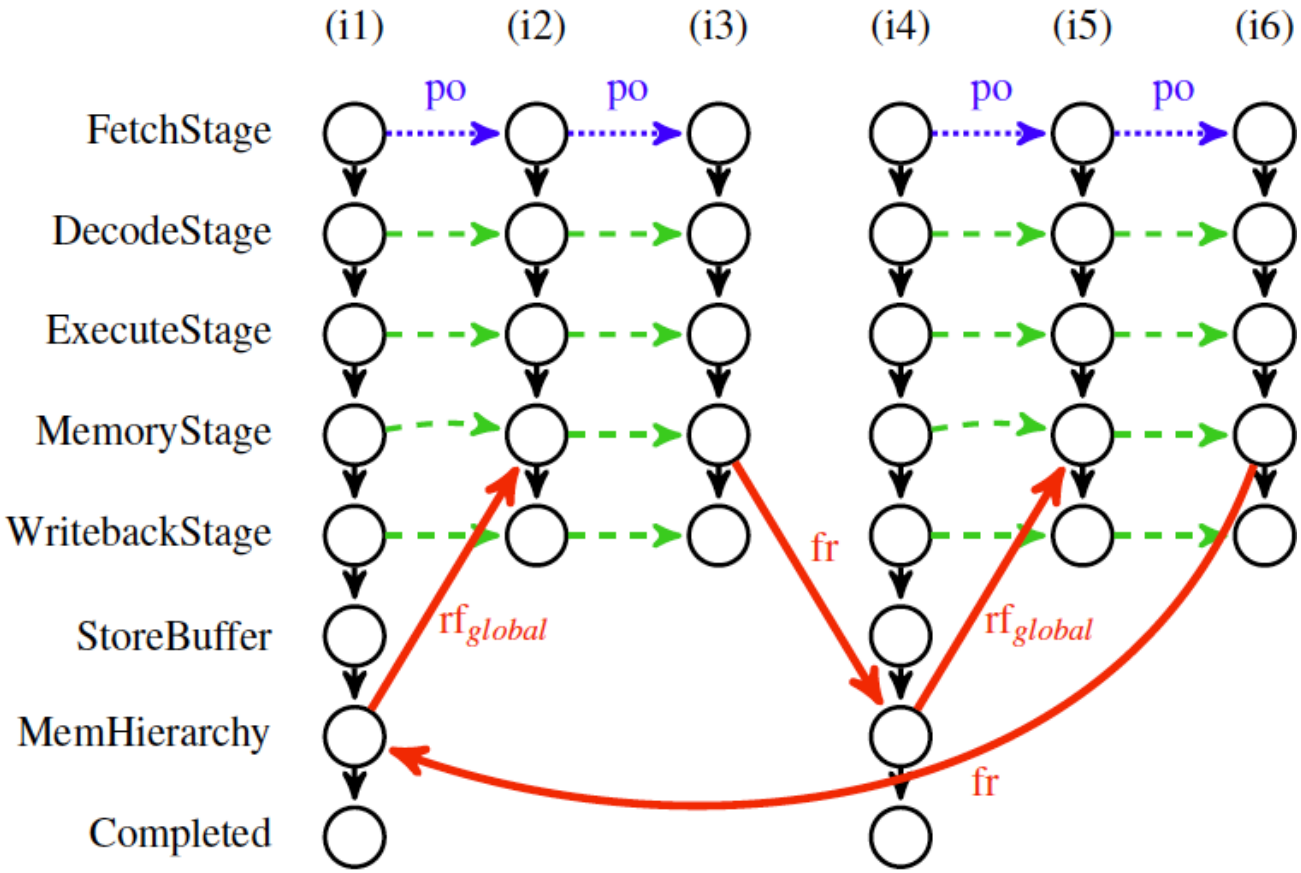


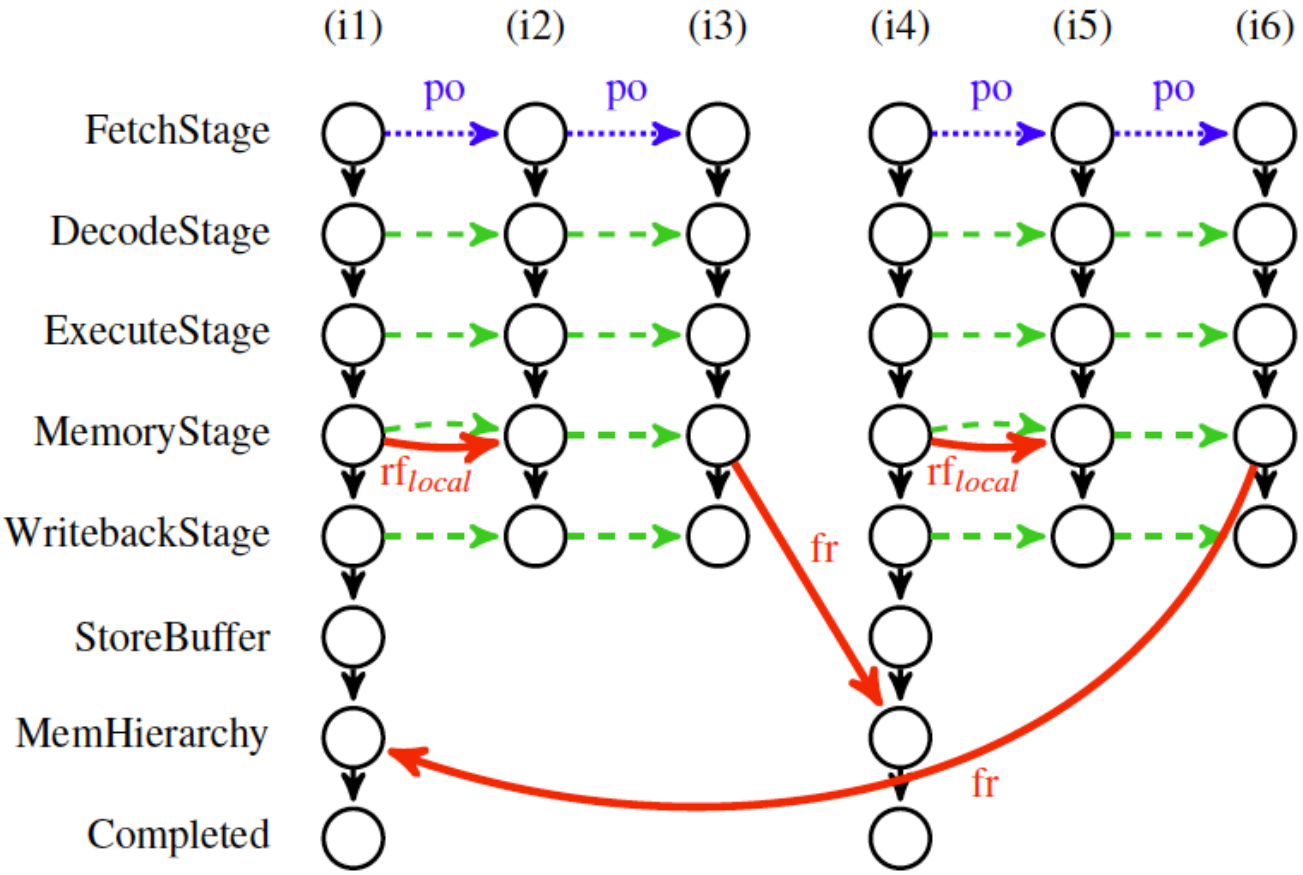
Reads from earlier ("from-reads")

At µarch Level, Cycle = Forbidden

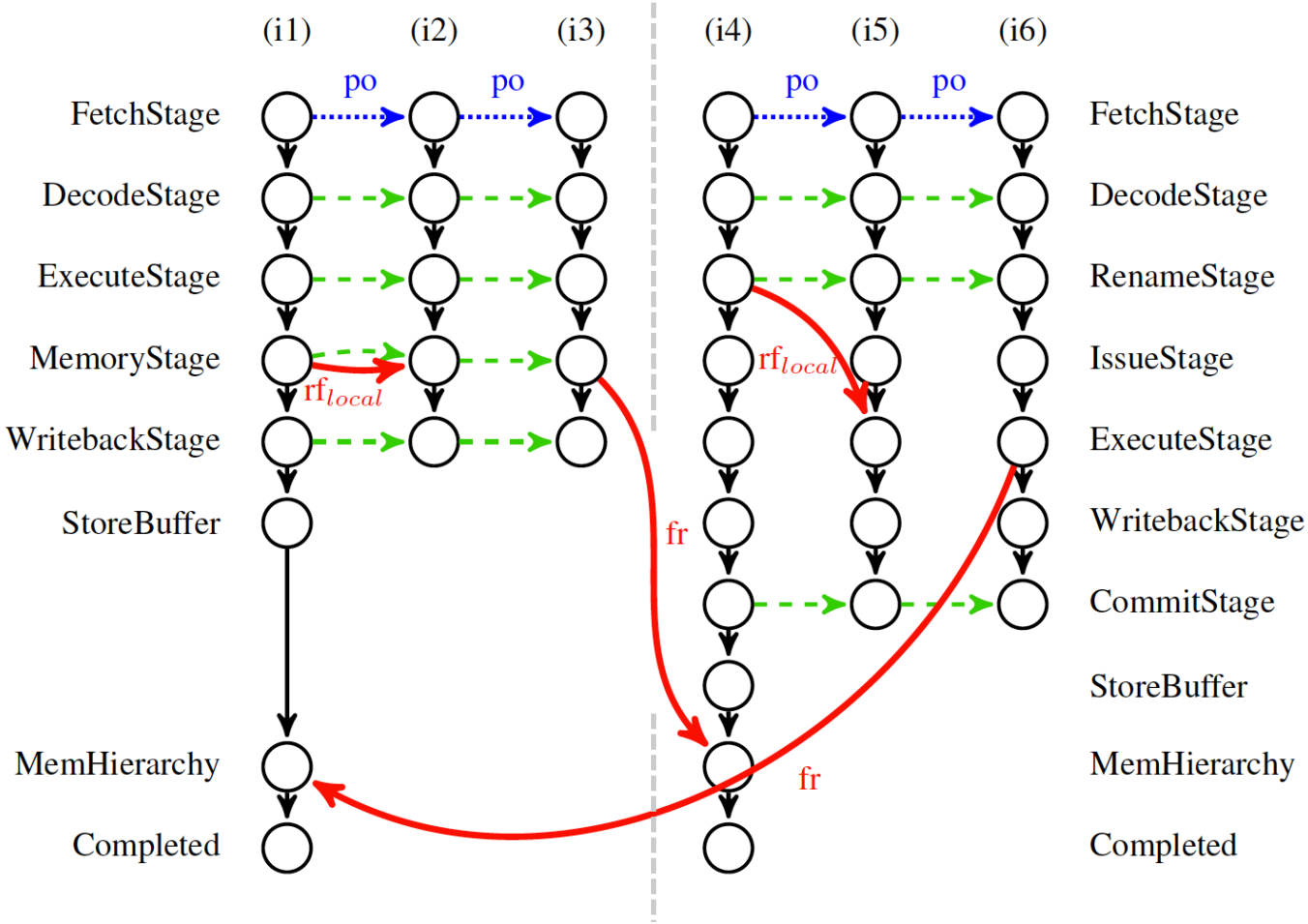


Satisfy reads from memory → µhb cycle → forbidden

At µarch Level, Cycle = Forbidden



Satisfy reads from store buffer → no µhb cycle → permitted

Supporting Heterogeneity

Different cores can have different µarches – no special support is needed